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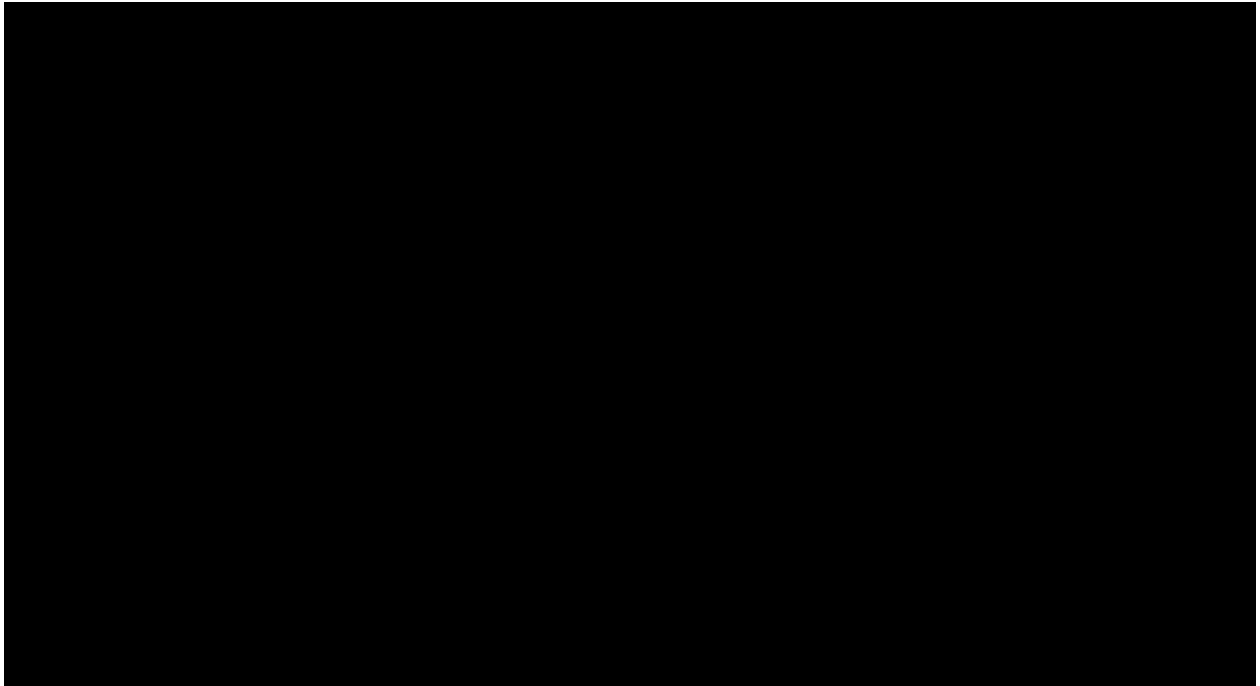
Areas of Interest (AOIs)

This Help Center article explains how to set 'Areas of Interest (AOIs)' for deriving eye-tracking statistics related to objects in iMotions 9.0 and later. It also explains how to work with 'Automated AOIs' released in iMotions 10.1.

The AOI editor is where you can analyze your eye-tracking data. AOIs are used for deriving eye-tracking metrics related to objects or other defined parts of the stimulus. This is done by drawing an 'area' around each of the objects, after which the algorithm returns the eye-tracking metrics related to these 'areas' (hence the objects). For instance, how long did it take before respondents first looked at each object? How long were the objects being looked at by respondents? And how many respondents looked at an object at least once?

Getting started tutorial

Let our Senior Customer Success Manager, Oscar Haven, walk you through how to analyze eye-tracking data with AOIs (video recorded in March 2021 and updated in September 2024).



An Overview of the User Interface

Enter the AOI editor

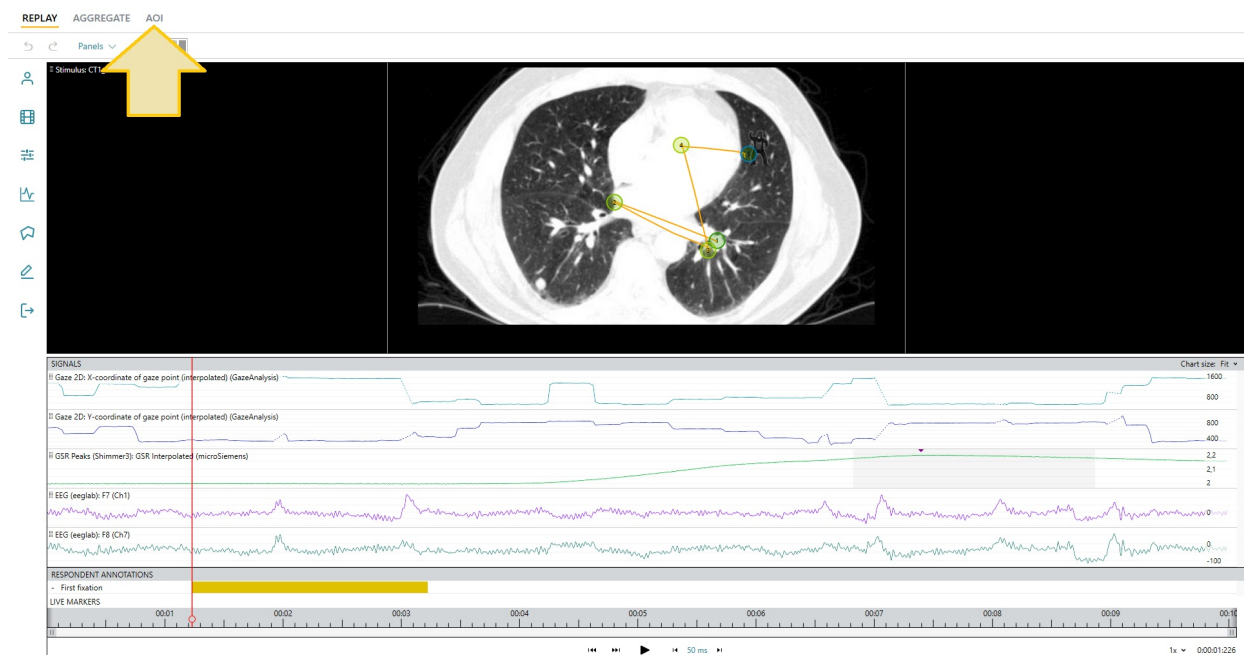
The AOI editor can only be accessed from the 'Analysis level'. This is the part of the software you enter by creating an analysis, or by clicking on the name of an existing analysis in the Library.

You can enter the AOI by using the shortcut in the analysis bar:

1. Click on the stimulus on which you want to define AOIs
2. Click on the option 'AOIs' in the analysis bar
3. Click 'Open analysis' to open the Analysis window on the AOI editor tab



Alternatively, if you are already working in the **Replay** or **Aggregate** editor, you can enter the AOI editor by clicking on the tab at the top of the editor. Please note that the editor will open the stimulus that you selected when you first opened the Analysis window and not the one that you last worked with.



What is in the UI?

In the AOI editor, you can find the following (*options and layout vary depending on the stimulus type and your customization*):

- Tabs to navigate between the different editors, i.e. the individual **Replay** editor, the **Aggregate** editor, and the **AOI** editor - *here highlighted with dark green box*
- A top bar with buttons to... - *here highlighted with a light green box*
 - undo and redo
 - choose the panels that you wish to see from a drop-down menu
 - re-arrange the panels in the UI
 - (if AOI is selected) view and modify the size and position of the AOI, as well as to add padding to it
 - a shortcut to this [Help Center article](#)
- A menu on the left-hand side (*here highlighted with red box*). Its different options are explained in the *next section*
- Blue buttons to... - *here highlighted with a yellow box*
 - start **drawing AOIs** manually
 - place an automated AOI with the help of computer vision (available from iMotions 10.1; requires the module for Automated AOIs)
 - **multiselect** existing AOIs
- A panel showing the currently selected stimulus. AOI shapes are shown here overlaid with the stimulus - *here highlighted with an orange box*
- A panel to display the **results** once they are calculated - *here highlighted with a purple box*
- A panel showing the **AOI timeline**, *only for dynamic stimuli (videos, screen recordings, ...)* - *here highlighted with a dark blue box*

- A panel showing Respondent or Stimulus Annotations, *only for dynamic stimuli (videos, screen recordings, ...)* - *here highlighted with a light blue box*
- Play controls to replay the stimulus, step forward/ backward, and zoom in and out of the timeline, *only for dynamic stimuli (videos, screen recordings, ...)* - *here highlighted with a brown box*

The screenshot displays two instances of the iMotions software interface. The top panel, titled 'CT4', shows a CT scan image of a human torso. The bottom panel, titled 'Basket_Gorilla', shows a video of a basketball game. Both panels have a 'REPLAY' tab selected, and the 'AOI' (Area of Interest) tab is active. In the 'Basket_Gorilla' panel, a blue box highlights the 'Draw AOI' button, and a brown box highlights the play controls at the bottom. The right side of the 'Basket_Gorilla' panel displays a table of metrics for three respondents: Gorilla, Pass, and Player.

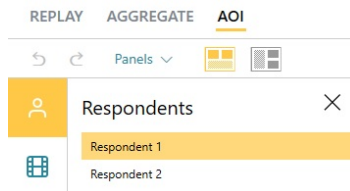
	Gorilla	Pass	Player
Information			
AOI duration (ms)	9729.7	8779	15500
AOI duration (%)	39.5	35.6	62.9
Size (cm2)	NA	NA	NA
Size (%)	NA	NA	NA
Respondent base	8	8	8
Fixation based metrics			
Respondent ratio (%)	100	100	100
Revisit count	3.4	3.8	11.6
Fixation count	9.9	6.4	38.9
TTFF AOI (ms)	2291.4	1745.4	581
Dwell time (ms)	3589.3	1975.5	10307
Dwell time (%)	36.9	22.5	66.5
First fixation duration (ms)	393.4	411.8	276.5
Saccade based metrics			
Respondent ratio (%)	100	62.5	100
Saccade count	5.6	2	23.3
Amplitude (deg)	2.4	1.3	1.8
Peak velocity (deg/s)	76.7	54.2	63.6

Below the table, there is a timeline visualization showing the sequence of events for the three respondents. The timeline includes a 'Gorilla' event (red bar), a 'Pass' event (green bar), and a 'Player' event (blue bar). The timeline is marked with time intervals: 00:05, 00:10, 00:15, and 00:20.

Options in the left-hand menu

Respondents

Here, you can select the Respondent whose data you wish to review and for whom you want to define AOIs.



Switching respondents is only available for the following stimulus types:

- Recordings from mobile eye-tracking glasses
- Recordings from a scene camera setup, including setups using a frame grabber
- Screen recordings
- Website recordings
- VR stimuli

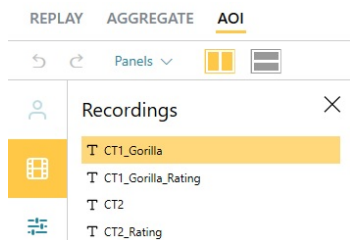
For all these stimulus types, every respondent has a timeline, because each respondent interacted with the stimulus, rather than passively watching the same stimulus on the computer screen.

Please note: For these recording types, you will also have to set AOIs individually per respondent. Read more about this under [Best practice tips](#)

For stimuli that either are static (images, survey slides; also [Gaze Maps](#) or [Scenes](#)) or where every respondent passively watched the same content (videos), jumping between respondents is not necessary. Since these stimuli can be [aggregated](#), the AOIs that you define are automatically applied across all respondents and you do not need to switch between respondents.

Recordings

Here, you can select the Stimulus for which you wish to define AOIs.

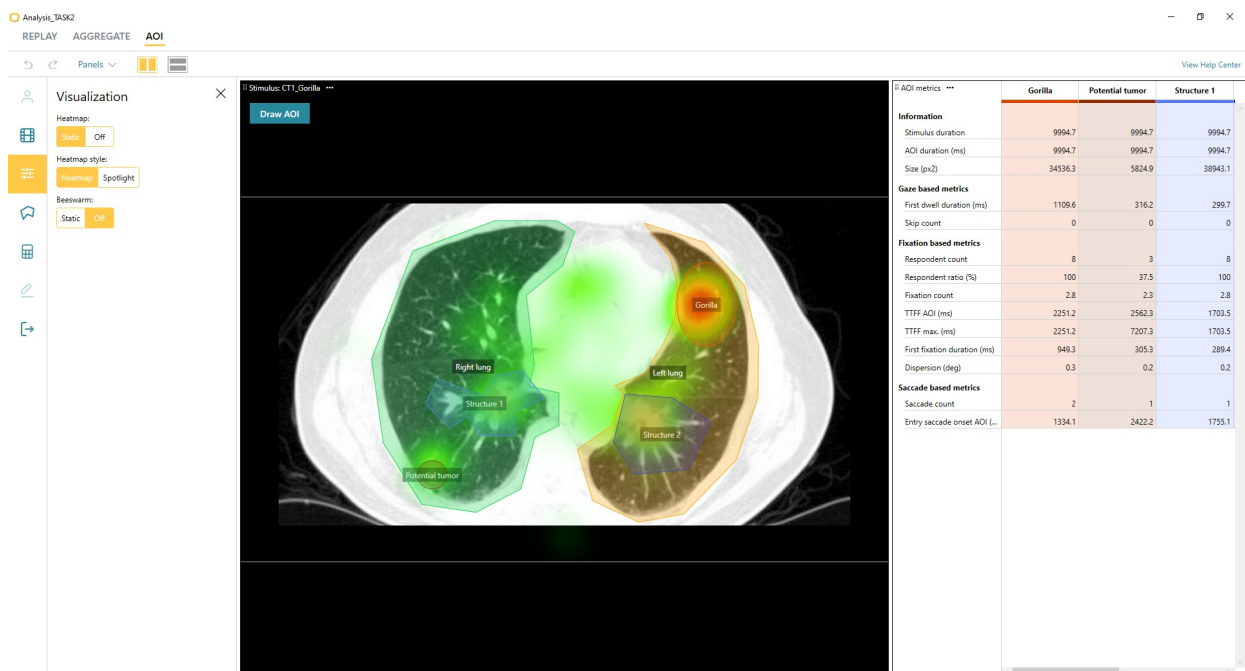


Visualization

Visualization options differ between stimuli that can be aggregated or not:

For stimuli that can be aggregated

Here, you can choose to overlay the stimulus with a heatmap.



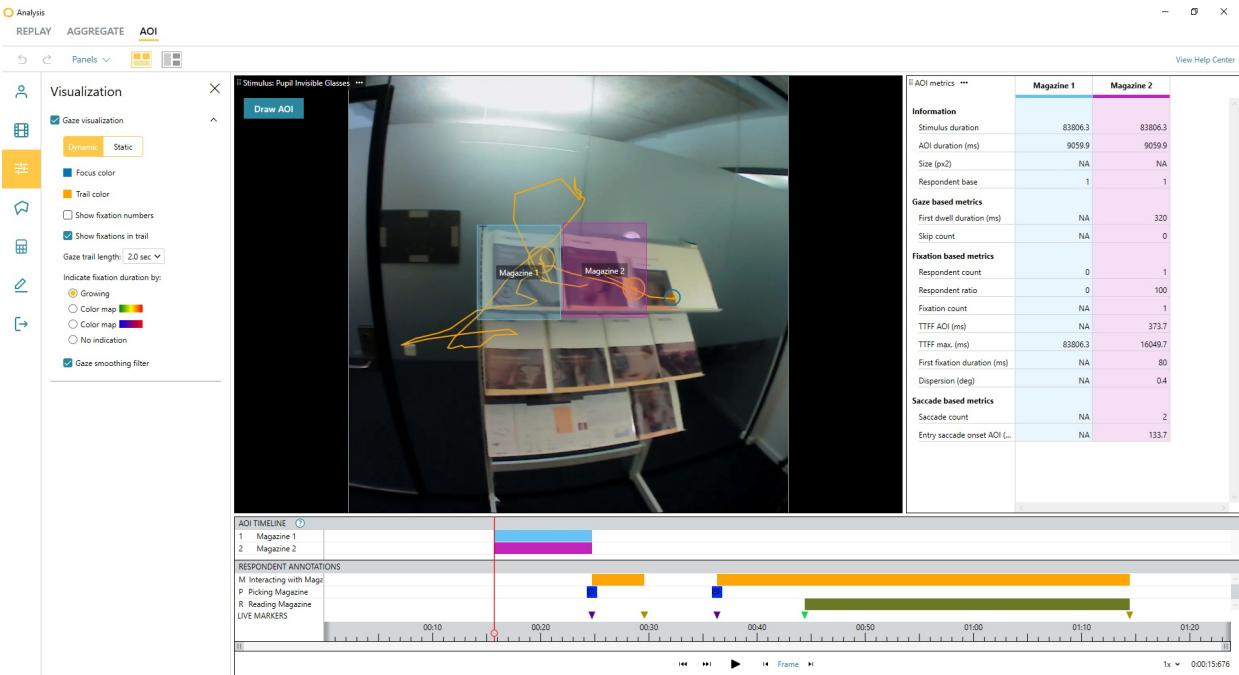
Please note: This is only possible for stimulus types that can be [aggregated](#), such as

- Images
- Survey slides

- Videos
- Gaze Maps
- Scenes

For stimuli that are individual recordings

Here, you can choose to replay the individual respondent's gaze overlaid with the stimulus

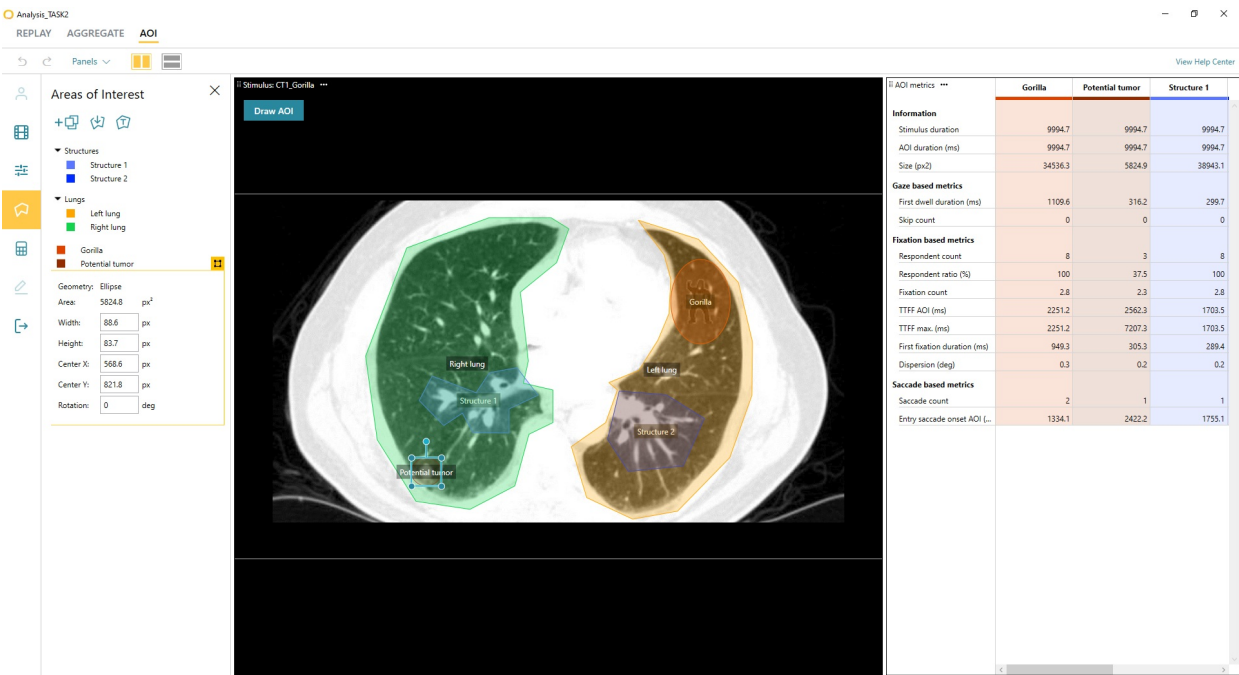


Please note: This is only possible for individual recordings per Respondent, such as:

- Recordings from mobile eye-tracking glasses
- Recordings from a scene camera setup, including setups using a frame grabber
- Screen recordings
- Website recordings
- VR stimuli

Areas of Interest

This part of the menu shows a list of all AOIs that are defined for the selected stimulus (and respondent). It also offers several options to interact with these AOIs:



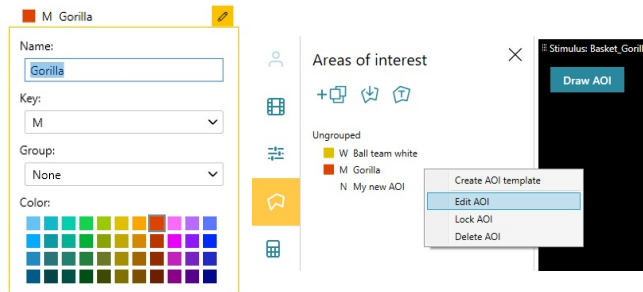
Change AOI name, color, or key

If you wish to

- edit the name of your AOI,
- change its color,
- or change/ assign a key to turn a dynamic AOI off or on,

you can

- click on the small pen icon  that appears when you hover over the AOI name with your mouse cursor
- or right-click on the AOI's name and select "Edit AOI".



These actions can be undone by using the arrow buttons or hitting Ctrl+Z on your keyboard.

Show/hide AOIs

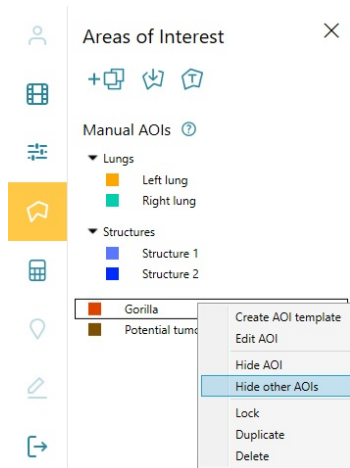
If you are working with many AOIs, it can help to hide some of them to have a less cluttered workspace and avoid accidentally clicking on or changing another AOI (please note that you can also [lock AOIs](#) to avoid accidental changes).

When an AOI is hidden, it is still available for computations and its metrics will be exported. If you wish to [delete an AOI](#) or [turn off a dynamic AOI](#), please read how to do this in the linked sections.

To hide or show an AOI from the workspace, click on its name in the Areas of Interest menu

	Potential tumor	Structure 1	Structure 2
Information			
Stimulus duration	9994.7	9994.7	9994.7
AOI duration (ms)	9994.7	9994.7	9994.7
Size (px2)	5824.9	38943.1	50038.1
Gaze based metrics			
First dwell duration (ms)	316.2	299.7	420.1
Skip count	0	0	0
Fixation based metrics			
Respondent count	3	8	10
Respondent ratio (%)	37.5	100	10
Fixation count	2.3	2.8	3.1
TFFF AOI (ms)	2562.3	1703.5	1442.1
TFFF max. (ms)	7207.3	1703.5	1442.1
First fixation duration (ms)	305.3	289.4	366.1
Dispersion (deg)	0.2	0.2	0.1
Saccade based metrics			
Saccade count	1	1	1
Entry saccade onset AOI (ms)	2422.2	1733.1	727.1

It is also possible to hide all other AOIs, keeping only the currently selected one.



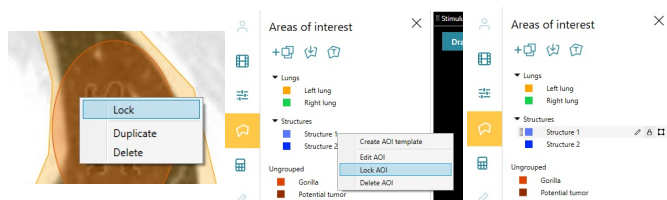
Lock AOI

You can prevent yourself from making accidental changes to your AOI by locking it.

To do so, you have three options:

- Right-click on the AOI shape and select "Lock",
- or right-click on their name in the **Areas of Interest** menu and select "Lock AOI",
- or click the little lock icon that appears when you hover over an AOI name in the **Areas of Interest** menu 

If you want to edit a locked AOI, you need to unlock it first, either by right-clicking on it or clicking the lock icon

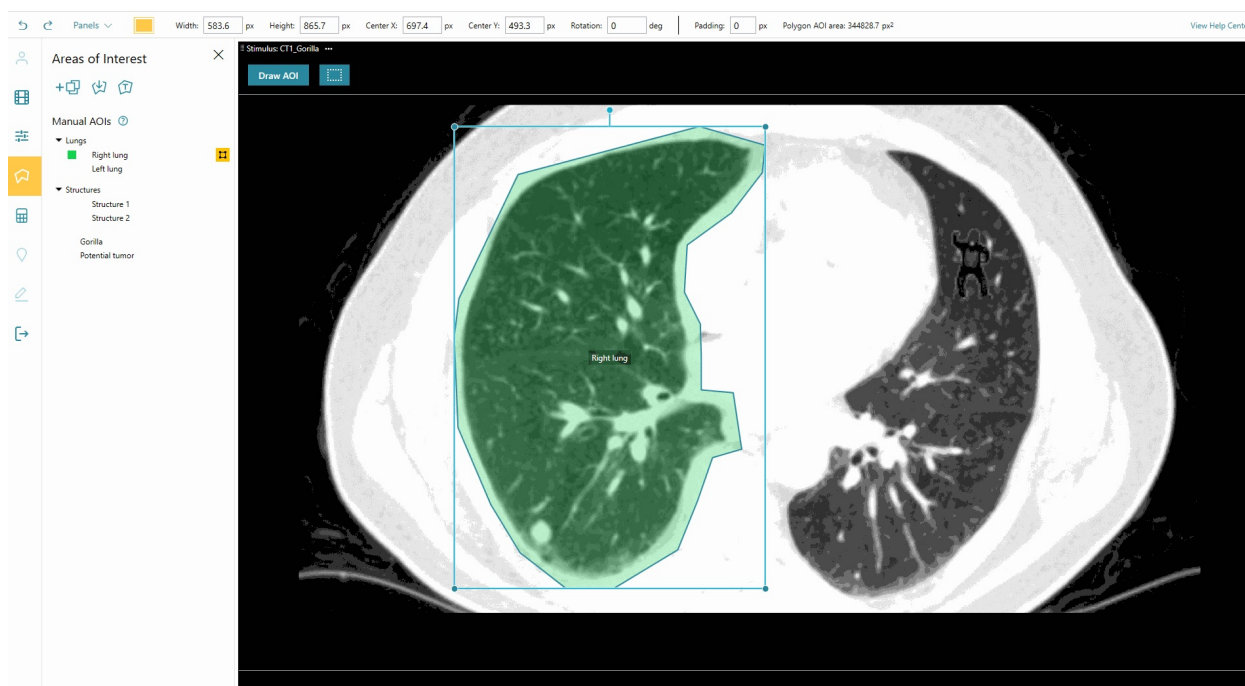


Locking an AOI can be undone by using the arrow buttons or hitting Ctrl+Z on your keyboard.

View and change AOI geometry

If you wish to view the size and position of a selected AOI in numeric values, you can do so by clicking on the AOI shape (with the **Areas of Interest** menu open) or clicking on the  icon next to the AOI name.

If you want to change the AOI shape with greater precision than just **dragging its boundaries**, you can enter exact values for the AOI width, height, coordinates, and rotation.

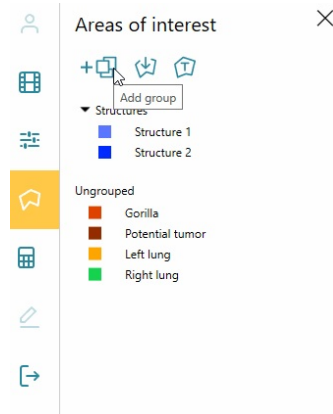


You can undo these actions by using the arrow buttons or hitting Ctrl+Z on your keyboard.

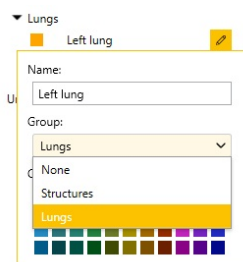
Grouping AOIs

If you have many AOIs that are associated with each other, you can group them. This helps to batch edit AOIs, but also to compute AOI metrics based on groups rather than individual AOIs.

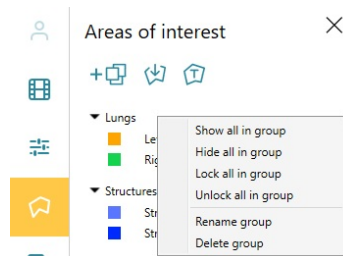
1. Click on the "Add group" icon
2. Enter the name of your new group
3. Drag and drop AOIs into the group



The group can also be assigned by editing the AOI:



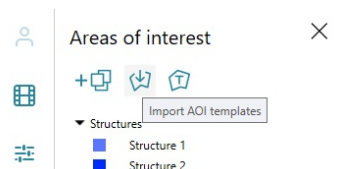
It is possible to batch-edit AOIs belonging to the same group by right-clicking on the Group name.



Working with AOI templates

Templates speed up your analysis because you can reuse one or many AOI shapes across stimuli and studies.

At the top of the Areas of Interest menu, you can find buttons to import and export AOI templates.



You can also create an AOI template by right-clicking on the name of an AOI.

REPLAY AGGREGATE **AOI**

AOI metrics

	Gorilla	Potential tumor	Structure 1	Structure 2	Left lung	Right lung
Information						
Stimulus duration	9994.7	9994.7	9994.7	9994.7	9994.7	99
AOI duration (ms)	9994.7	9994.7	9994.7	9994.7	9994.7	99
Size (px2)	34536.3	5824.9	38943.1	50038.9	256884.9	3591
Gaze based metrics						
Dwell count	2.4	2.7	2.6	3.5	4	
Revisit count	1.4	1.7	1.6	2.5	3	
Dwell time (ms)	2516.5	1282.2	938.7	1488.1	4814.2	27
Dwell time (%)	25.2	12.8	9.4	14.9	48.2	
Saccade based metrics						
Saccade count	2	1	1	1.6	6	
Entry saccade onset AOI (...)	1334.1	2422.2	1755.1	727.7	1028	
Amplitude (deg)	1.4	1.4	1.1	1.1	2.9	
Direction (deg)	175.2	286	203.7	163.1	228	2

By hovering over the  icon, you can read a definition of the metric and learn more about it.

REPLAY AGGREGATE **AOI**

AOI metrics

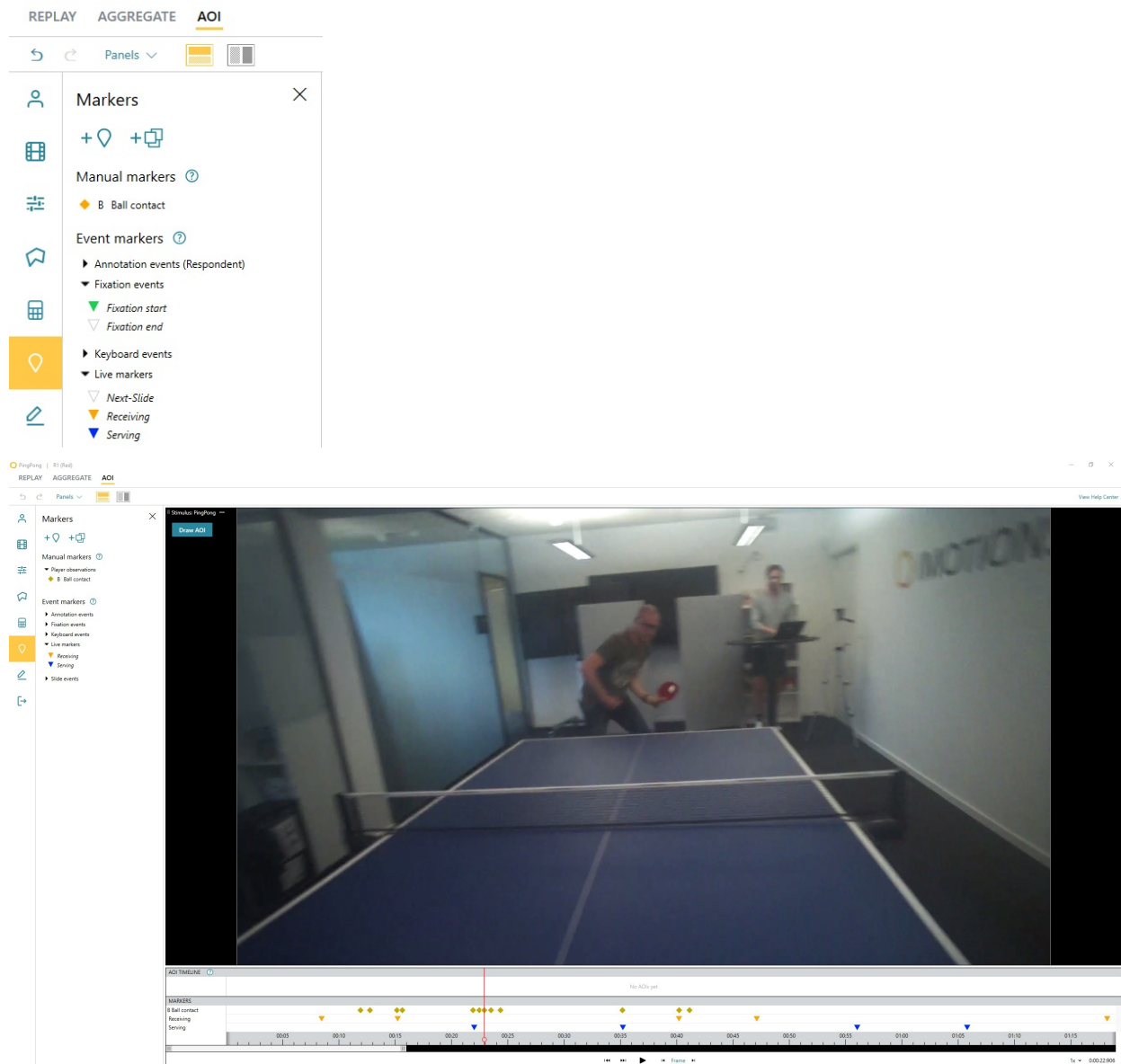
	Gorilla	Potential tumor	Structure 1	Structure 2	Left lung	Right lung
Information						
Stimulus duration	9994.7	9994.7	9994.7	9994.7	9994.7	99
AOI duration (ms)	9994.7	9994.7	9994.7	9994.7	9994.7	99
Size (px2)	34536.3	5824.9	38943.1	50038.9	256884.9	3591
Valid data	97.7	97.7	97.7	97.7	97.7	
Respondent base	8	8	8	8	8	
Fixation based metrics						
Respondent count	8	3	8	8	8	
Respondent ratio (%)	100	37.5	100	100	100	
Revisit count	1.3	1.3	1.6	2.5	3	
TTFF AOI (ms)	2251.2	2562.3	1703.5	1442.8	945.3	5
Dwell time (ms)	2422.5	1431.9	872.3	1393	4439.6	23
Dwells with fixations	24.2	14.3	8.7	13.9	44.4	
First fixation duration (ms)	972.9	554.1	329.3	380.8	541.5	2
First fixation duration (ms)	949.3	305.3	289.4	366.9	462.6	1
Saccade based metrics						
Respondent count	4	2	3	5	8	
Respondent ratio (%)	50	25	37.5	62.5	100	
Amplitude (deg)	1.4	1.4	1.1	1.1	2.9	
Direction (deg)	175.2	286	203.7	163.1	228	2

Dwells with fixations
Average of how often the respondents looked at the AOI and fixated on it at least once.

Markers

Markers mark a point in time. There are two kinds of markers:

- Event markers were recorded during the data collection (such as [Live Markers](#), keypresses, mouse clicks, [API events](#), etc.) or generated from recorded sensor data (such as fixations from eye tracking data)
- Manual markers can be generated during the analysis



Event markers cannot be changed. It is possible to create a duplicate of an Event marker, which then can be edited the same way as a Manual marker.

Markers are only available for individual recordings, such as

- Recordings from mobile eye-tracking glasses
- Recordings from a scene camera setup, including setups using a frame grabber
- Screen recordings
- Website recordings
- VR stimuli

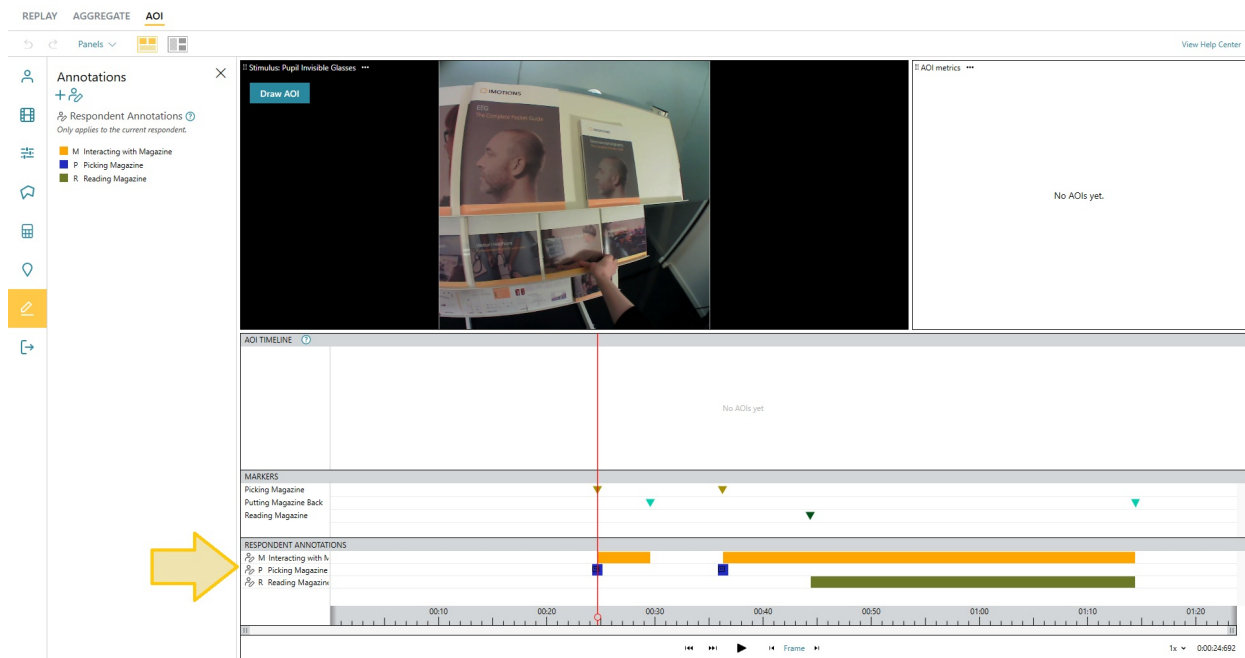
For more information on how to set and edit Markers, please refer to the article about the [Replay](#).

Annotations

Annotations are a tool to mark up the time ranges of your data. There are two types of Annotations:

- Respondent Annotations, also available in the [individual Replay editor](#)
- Stimulus Annotations, also available in the [Aggregate Replay editor](#)

In addition to Annotations, there are also [point markers](#).



Annotations can be viewed in their panel next to the AOI timeline (you can choose to show or hide the Annotation panel in the [interface options](#)).

Viewing Annotations in the AOI editor is only available for dynamic stimuli, i.e.

- Videos
- Recordings from mobile eye-tracking glasses
- Recordings from a scene camera setup, including setups using a frame grabber
- Screen recordings
- Website recordings
- VR stimuli

For static stimuli (Images, Survey slides, [Gaze Maps](#), [Scenes](#)), please refer to the [Replay](#) or [Aggregate](#) editors if you wish to set Annotations.

If your stimulus can be aggregated (Videos), you will see Stimulus Annotations in the AOI editor, which are synchronized with the [Aggregate](#) editor.

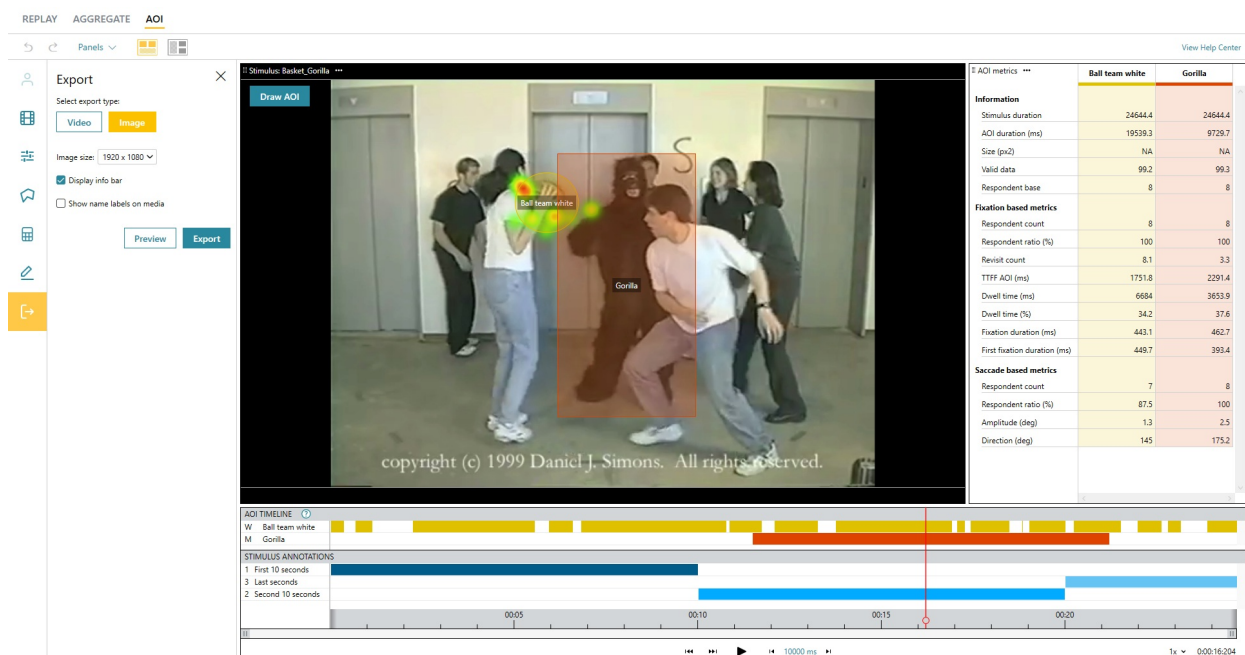
If your stimulus is an individual recording that you review target=" _blank" rel="noopener">Respondent basis, you will see Respondent Annotations in the AOI editor, which is synchronized with the [Replay](#) editor.

Export

If you would like to share visualizations of your AOIs and AOI metrics with colleagues or clients or use them in a report or publication, you can export screenshots of the workspace.

You can choose to export a still image of the workspace by choosing "Image".

- The option to "Display info bar" adds a bottom bar with information about the current stimulus
- The option to "Show name labels on media" includes the panel labels in the screenshot



You can also choose to export a video of your recording along with the AOI metrics.

- You have the option to either export the entire recording duration or just a selected time range
- You can choose how much you wish to zoom into the timeline
- The option to "Follow cursor" changes whether the timeline remains static and the red timeline cursor moves forward, or whether you wish the cursor to remain static and the timeline to move
- You can customize the quality and format of your exported video

Export

Select export type:

Video Image

Video options

Export size: 1920 x 1080

Show info bar in video

Show name labels on media

Time range: Full Custom

Start time: 00:00:11:499

End time: 00:00:21:225

Tip: you can select a custom range by holding the Ctrl key while dragging over the timeline.

Time zoom: Entire timeline

Follow cursor

Frames per second: 30

Video format: mp4

Export video

AOI metrics

	Ball team white	Gorilla
Information		
Stimulus duration	24644.4	24644.4
AOI duration (ms)	19539.3	9729.7
Size (px2)	NA	NA
Valid data	99.2	99.3
Respondent base	8	8
Fixation based metrics		
Respondent count	8	8
Respondent ratio (%)	100	100
Revisit count	8.1	3.3
TTFP AOI (ms)	1751.8	2291.4
Dwell time (ms)	6684	3653.9
Dwell time (%)	34.2	37.6
Fixation duration (ms)	443.1	462.7
First fixation duration (ms)	448.7	393.4
Saccade based metrics		
Respondent count	7	8
Respondent ratio (%)	87.5	100
Amplitude (deg)	1.3	2.5
Direction (deg)	145	175.2

AOI TIMELINE

W Ball team white

M Gorilla

STIMULUS ANNOTATIONS

1 First 10 seconds

2 Last seconds

3 Second 10 seconds

00:05 00:10 00:15 00:20

1x 00:16:204

Exporting AOI Metrics

To export your AOI metrics into CSV files, use the Analysis Export process that is described here (additional information can be found in [this Help Center article](#) about the Analysis Export interface).

1. Select your analysis that includes calculated AOI metrics
2. Clicking the Export button
3. Select the R Processing tab
4. Select AOI Metrics Export (indicated by the turquoise button to the left of the panel)
5. (Optional) Use the drop-down button (on the right of the panel) for export settings.

Define AOIs

Draw AOIs

Basic workflow to draw an AOI

1. Click on "Draw AOI" at the top left side of the stimulus
2. Enter a name for your AOI (*optional; the name can be edited later*)
3. Assign a key with which you can activate/ deactivate your AOI (*optional; only available for dynamic stimuli. Read more about this in the section on how to set dynamic AOIs*)
4. Choose the kind of shape that you want to use for your AOI (*for more details, see next paragraphs*)
5. Assign a color for your AOI (*optional; the color can be edited later*)
6. Click on "Start drawing"
7. Click and drag on the stimulus to place a rectangular or circular AOI. Click on the stimulus for every corner of the polygonal AOI you want to place.
8. If you change your mind, you can click "Cancel Drawing" at the top left side of the stimulus

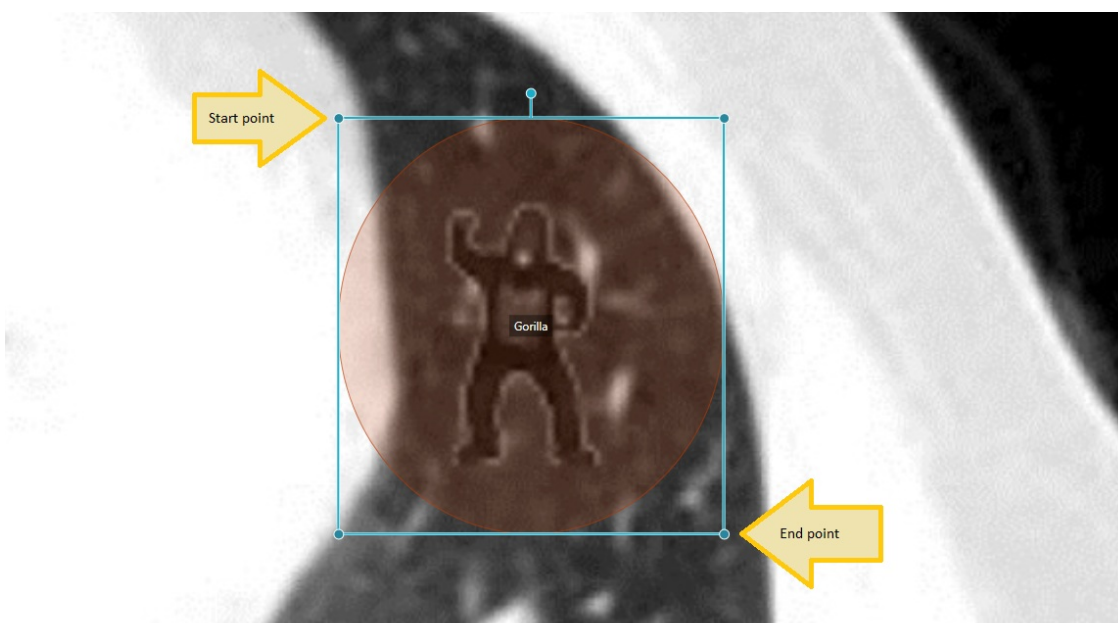


Drawing a rectangular AOI

1. After clicking on the "Start drawing" button (see [Basic workflow to draw an AOI](#)), the mouse cursor turns into a crosshair to indicate that you are in drawing mode
2. Place the AOI by *clicking* on one corner of the object you want to draw the AOI over
3. *Hold* the mouse pressed and *drag* the cursor over the object. The AOI will dynamically follow your cursor
4. *Release* the mouse when the rectangle has its desired size

Drawing an elliptical AOI

Drawing a circular AOI involves the same steps as drawing a rectangular AOI. Imagine that the ellipse is placed inside a surrounding rectangle to find an adequate starting point.



Drawing an AOI with a variable number of points (polygon)

This tool allows you to draw any shape beyond rectangles and ellipses. You are free to choose the amount of points, but we recommend not to set too detailed AOIs to keep the readjusting effort low and keep a margin around the relevant object (see also: [Best practice tips](#))

1. After clicking on the "Start drawing" button (see [Basic workflow to draw an AOI](#)), the mouse cursor turns into a crosshair to indicate that you are in drawing mode
2. *Click* on the location for the first point of your AOI
3. Release the mouse, move the cursor, and click at the location of the next point
4. Repeat step 3 for every point you want to draw. All defined points will be highlighted with a white circle. It is not possible to change a point while you are in drawing mode, but their location can be adjusted later. Please note that the number of points cannot be changed once the AOI has been created
5. To close the shape and end drawing mode, click again on the first point



Tip: Automated placement of AOIs

For static as well as [dynamic stimuli](#), it is possible to have [AOIs be placed automatically](#) through computer vision rather than drawing them manually.



Duplicate AOIs / Use templates

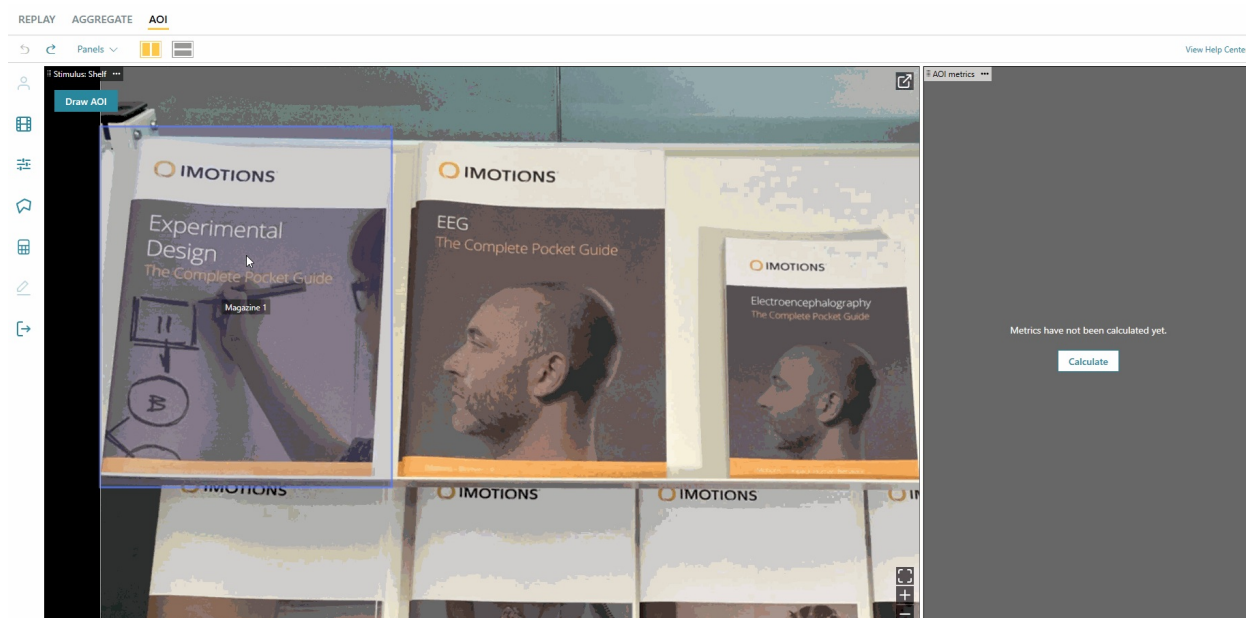
You can save time during your analysis by reusing AOIs that you have already created.

Option 1: Duplicating an AOI

You can duplicate an AOI by right-clicking on the AOI shape and selecting "Duplicate"



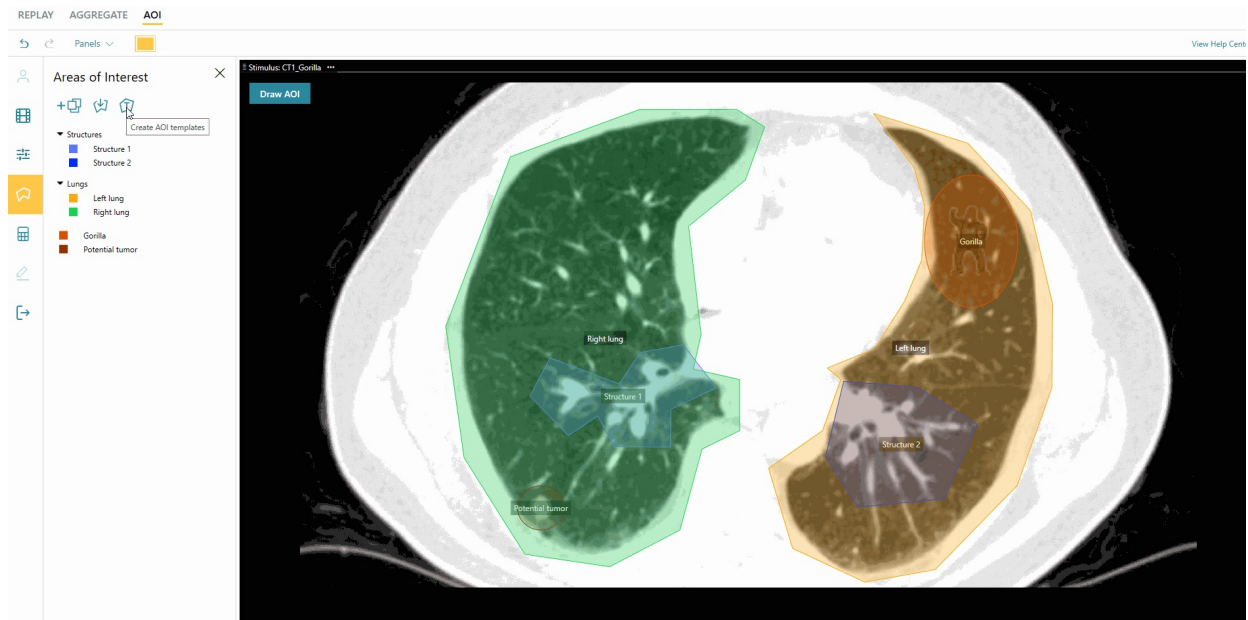
Alternatively, you can create a copy of your AOI by holding the CTRL key and dragging an AOI.



Option 2: Working with templates

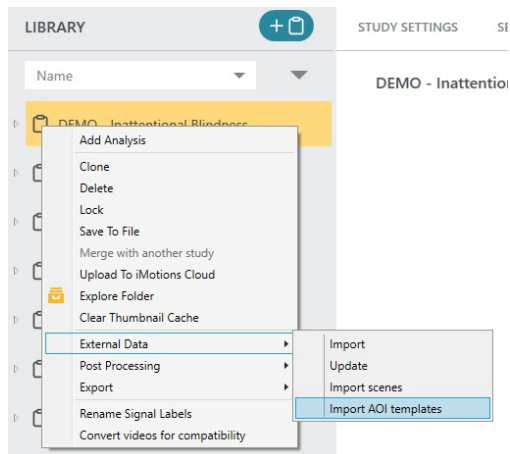
If you wish to reuse an AOI in a different stimulus, you can create AOI templates in the [Areas of Interest](#) menu

For easier management of your AOIs, you can choose to save all AOIs that belong to the same stimulus or study into the same folder. To do so, you can change the Destination path and save all AOIs into a new folder.

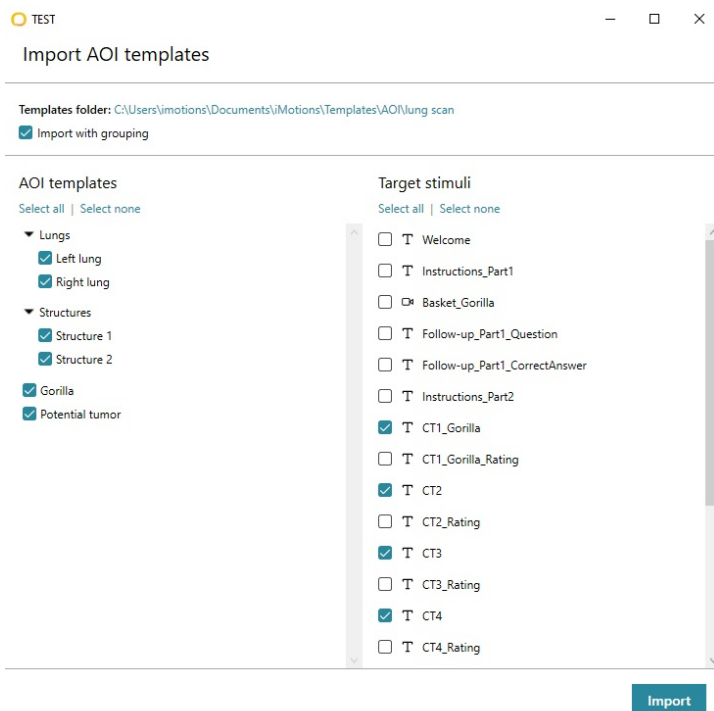


Batch application of AOI templates

If you wish to efficiently apply the same AOI template to many stimuli at once, you can do so by selecting External Data > Import AOI templates after clicking on the study name.



You can then select the AOIs that you want to batch apply to which stimuli.



Draw AOI Grids

The AOI Grid Tool (available after clicking 'Draw AOI') is used to draw a rectangular grid of AOIs, with a selected number of rows and columns. The grid cannot be changed once it is set and will always be active on the stimulus it was placed on. You can choose the number of rows and columns of your grid, up to a maximum of 20.



Import from CSV

It is possible to define static or dynamic AOIs (i.e. their shape, start and end times, and other properties) in an external CSV file and then import them to your study.

This CSV file needs to contain the following columns:

- **Respondent Name**
 - Allowed values are: Text input (or blank)
 - The column can only be left blank if it constitutes a keyframe succeeding an initial dynamic AOI definition row
 - For static AOI definitions, the name should correspond to one valid respondent across the file (as a static AOI will be applied for all respondents) compared to a dynamic AOI which is unique for each respondent
 - Is case sensitive
- **Stimulus Name**
 - Allowed values are: Text input (or blank)
 - The column can only be left blank if it constitutes a keyframe succeeding an initial dynamic AOI definition row

- Is case sensitive
- *AOI Name*
 - Allowed values are: Text input (or blank)
 - The column can only be left blank if it constitutes a keyframe succeeding an initial dynamic AOI definition row
- *Color*
 - Allowed values are: Text input or blank
 - Optional
 - Must be specified in hex code
- *Group*
 - Allowed values are: Text input or blank
 - Optional
 - Includes the AOI within the named group
- *Timestamp (ms)*
 - Allowed values are: Text input
 - Allowed values are integers or floats (numeric values with decimals indicated by a ".").
 - Start time of the AOI relative to stimulus start (Video or Screen Recording), in milliseconds
 - For a static AOI definition, the timestamp is ignored (but still needs to be set, e.g. to "0")
- *Is active*
 - Allowed values are: Text input (dynamic AOI) or blank (static AOI)
 - A value of "1" will activate the AOI
 - A value of "0" will de-activate the AOI
- *Points*
 - Allowed values are a series of value pairs, X and Y coordinates - either as integers or floats (numeric values with decimals indicated by a ".").
 - X and Y values are separated by "," and pairs using ";"
 - Coordinates that make up an AOI shape (polygon)
 - Minimum of 3 points must be defined
- *Translation*
 - Allowed values are value pairs, X and Y offsets - either as integers or floats (numeric values with decimals indicated by a ".").
 - X and Y values are separated by ";"
 - Translation effectively means offsetting the shape with the provided coordinate values
- *Scale*
 - Allowed values are value pairs, width and height (AOI) scaling factors - either as integers or floats (numeric values with decimals indicated by a ".").
 - Width and height values are separated by ";"
- *Rotation*
 - Allowed values are: Text input
 - Values are provided in Degrees - either as integers or floats (numeric values with decimals indicated by a ".").
- *Interpolate*
 - Allowed values are: Text input (dynamic AOI) or blank (static AOI)
 - A value of "1" will enable interpolation to the next key-frame
 - A value of "0" will disable interpolation to the next key-frame

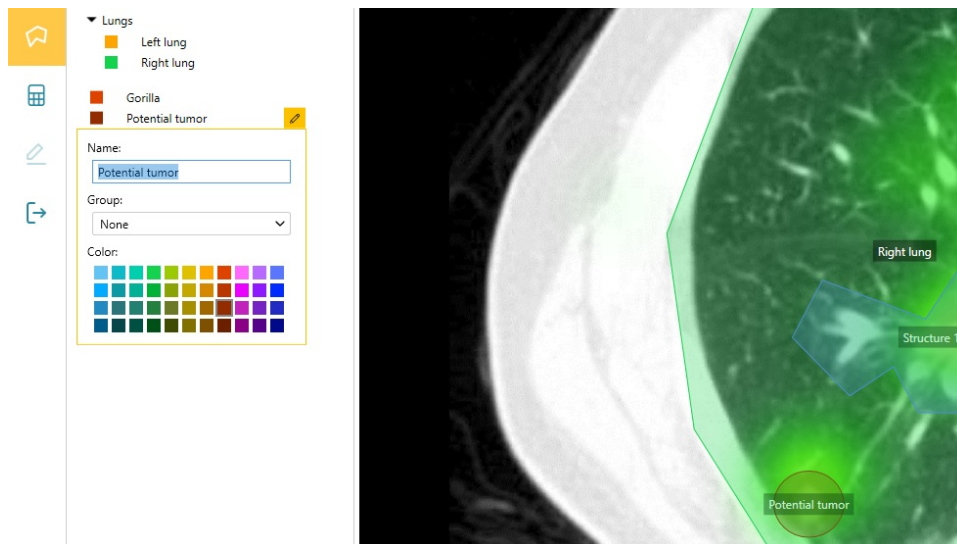
Download sample [static AOI](#) or [dynamic AOI](#) file.

Edit AOIs

Change AOI name or color

To rename your AOI or change its color, open the [Areas of Interest](#) menu 

1. Hover over the name of the AOI and then click on the small pen icon  or right-click on the AOI name and select "Edit AOI"
2. Enter the new AOI name in the dialogue
3. Click on a new color
4. For dynamic AOIs: Enter a key that you can turn the AOI off or on with



You can undo these actions by using the arrow buttons or hitting Ctrl+Z on your keyboard.

Relocate AOI

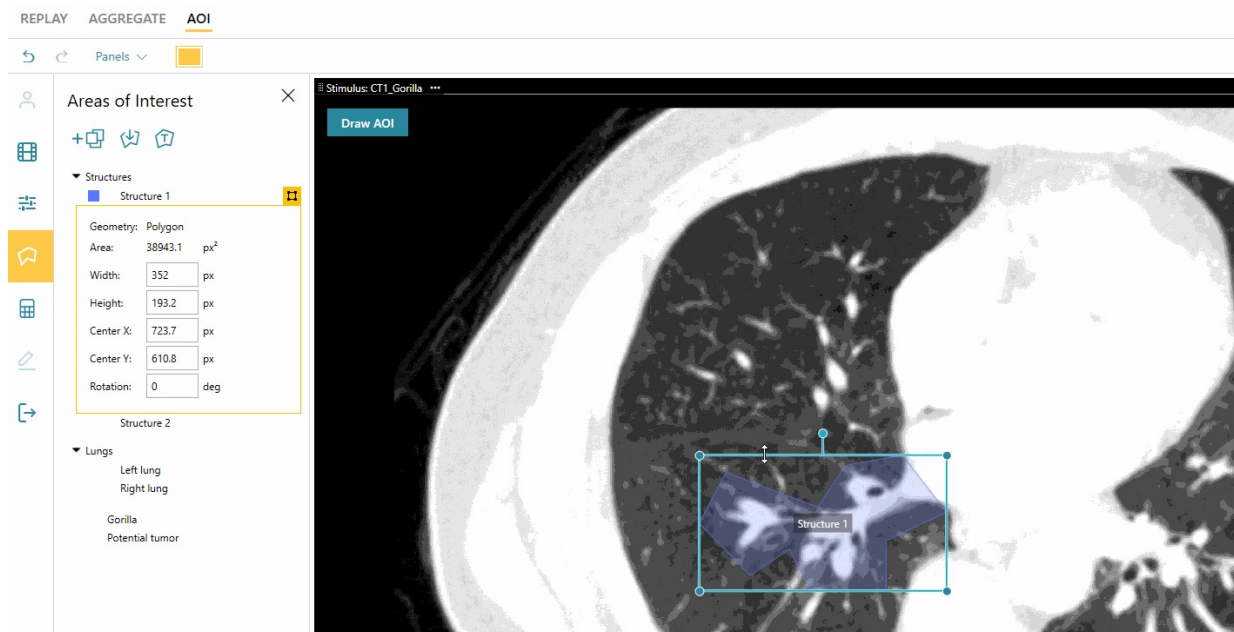
Option 1: You can change the location of your AOI by clicking on the shape and dragging it to a new location

Tip: If you want to move the AOI in a strict horizontal or vertical direction, you can hold the Shift key while dragging the AOI

Option 2: You can move a selected AOI with the arrow keys while holding the Ctrl key pressed.

Option 3: Manually enter new coordinates for the AOI's center point

1. Open the **Areas of Interest** menu
2. Hover over the name of the AOI and then click on the small bounding box icon or click on the AOI shape
3. Manually enter new values for the x and y coordinates of the AOI's center point



You can undo these actions by using the arrow buttons or hitting Ctrl+Z on your keyboard.

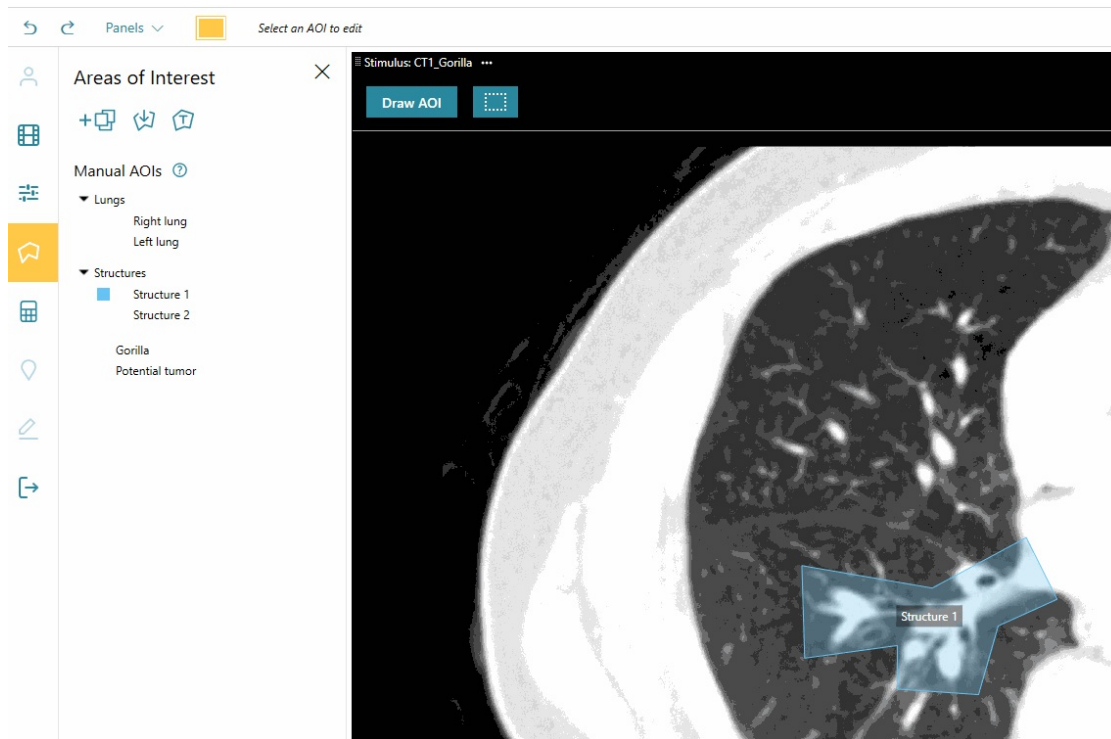
Resize AOI

Option 1: Change the size of the AOI by dragging the borders of its bounding box.

1. A single click on the AOI will display the bounding box
2. Drag the borders of the bounding box to change the width or height of the AOI or
3. Drag a corner of the bounding box to change the width and height of the AOI simultaneously

Option 2: Change the size of the AOI by manually entering width and height values

1. Open the **Areas of Interest** menu
2. Hover over the name of the AOI and then click on the small bounding box icon or click on the AOI shape
3. Manually enter new values for the width and height



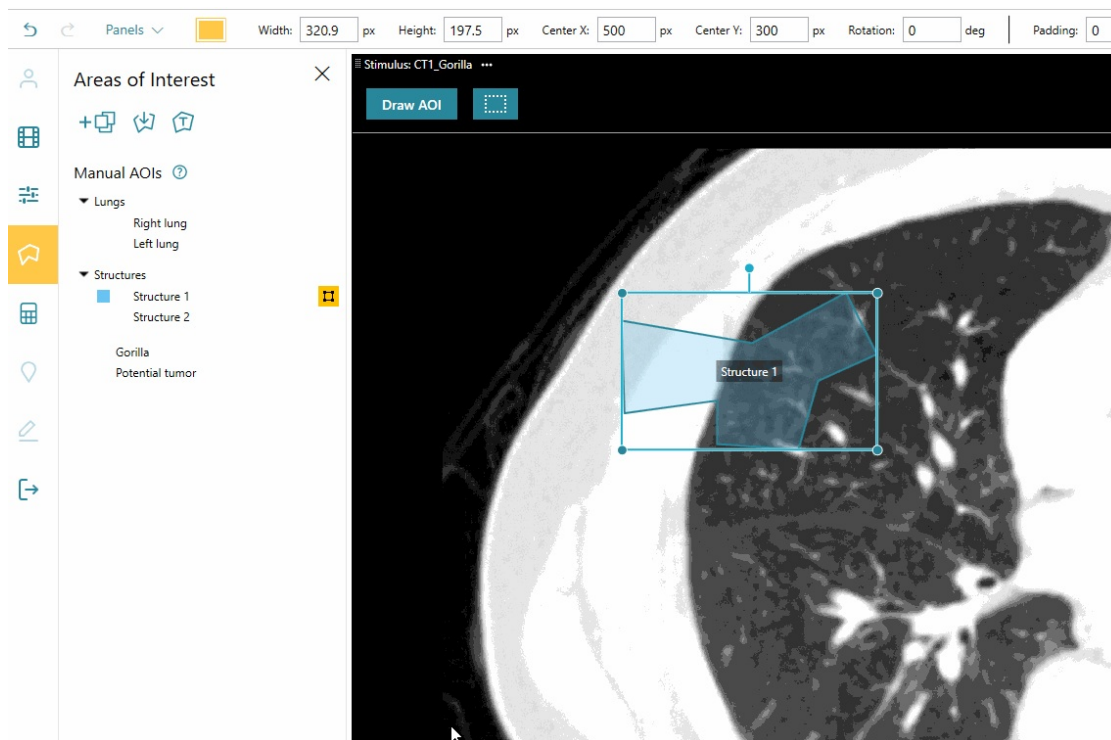
You can undo these actions by using the arrow buttons or hitting Ctrl+Z on your keyboard.

Rotate AOI

Option 1: Drag the handle of the bounding box to rotate the AOI

Option 2: Change the rotation of the AOI by entering a value manually

1. Open the **Areas of Interest** menu
2. Hover over the name of the AOI and then click on the small bounding box icon  or click on the AOI shape
3. Manually enter a new value for the rotation

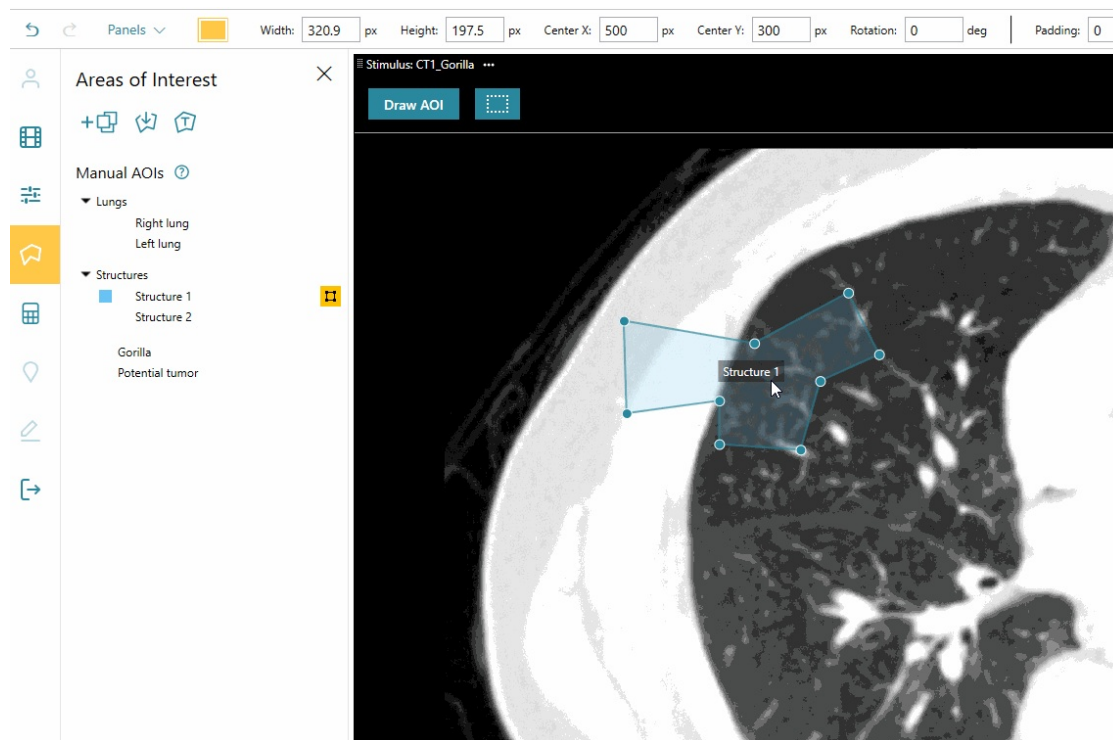


Rotating an AOI can be undone by using the arrow buttons or hitting Ctrl+Z on your keyboard.

Reshape polygonal AOI

In addition to changing the overall width and height of the polygon or rotating it, you can also move the individual corner points

1. Click on the AOI two times to go into point editing mode, which will display circles over each point
2. Drag the point of your choice to its new location



You can undo these actions by using the arrow buttons or hitting Ctrl+Z on your keyboard.

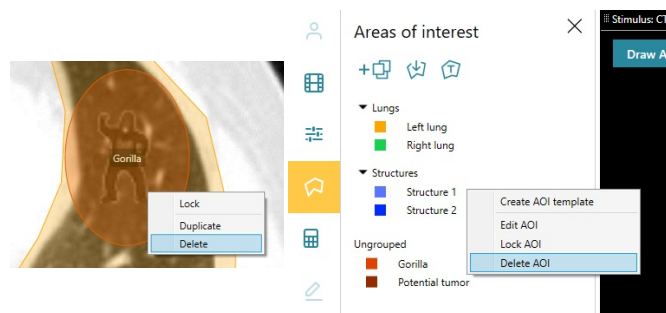
Please note:

- Clicking on an AOI once displays the bounding box, allowing you to edit the width and height of the overall AOI
- For polygons, a second click will display circles over the individual points, allowing you to edit the shape in more specific detail
- You cannot add or delete corner points. The amount of corner points will always remain the same

Delete AOI

AOIs can be deleted by

- right-clicking on the AOI shape and selecting "Delete",
- or by right-clicking on their name in the **Areas of Interest** menu and selecting "Delete AOI"



Deleting an AOI can be undone by using the arrow buttons or hitting Ctrl+Z on your keyboard.

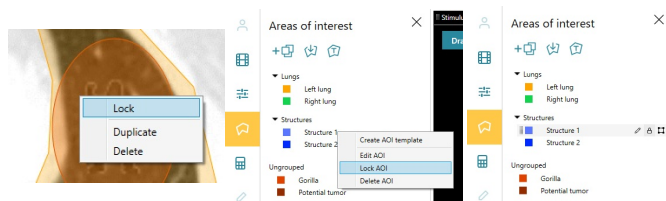
Lock AOIs

You can prevent yourself from making accidental changes to your AOI by locking it.

To do so, you have three options:

- Right-click on the AOI shape and select "Lock",
- or right-click on their name in the **Areas of Interest** menu and select "Lock AOI",
- or click the little lock icon that appears when you hover over an AOI name in the **Areas of Interest** menu

If you want to edit a locked AOI, you need to unlock it first, either by right-clicking on it or clicking the lock icon



Locking an AOI can be undone by using the arrow buttons or hitting Ctrl+Z on your keyboard.

Multiselecting AOIs

It is possible to select multiple AOIs at once so that they can be moved, resized, or rotated together.



Working with dynamic AOIs

If you have a static stimulus (like an image or survey slide, but also Gaze Maps or website exports), defining AOIs is very easy because the relevant objects remain where they are. If you have a dynamic stimulus, however (like a video, a mobile glasses recording, a screen recording, or a website interaction), then defining AOIs gets more complex.

This is because the objects change location (e.g. because of scrolling or because the respondent was walking), and disappear as well as reappear on the screen (e.g. because of cuts in the video or because the respondent turns their head). In this case, you have to ensure that the AOI is always over the object in the stimulus.

You can do so by using **Automated AOIs** with inbuilt object tracking (available from iMotions 10.1; requires module for Automated AOIs) or manually editing your AOI so that it is always following the object. (Read more about this under [Best practice tips](#)).

Before introducing how to implement AOIs on dynamic stimuli with either way of working, we will introduce some basic terms:

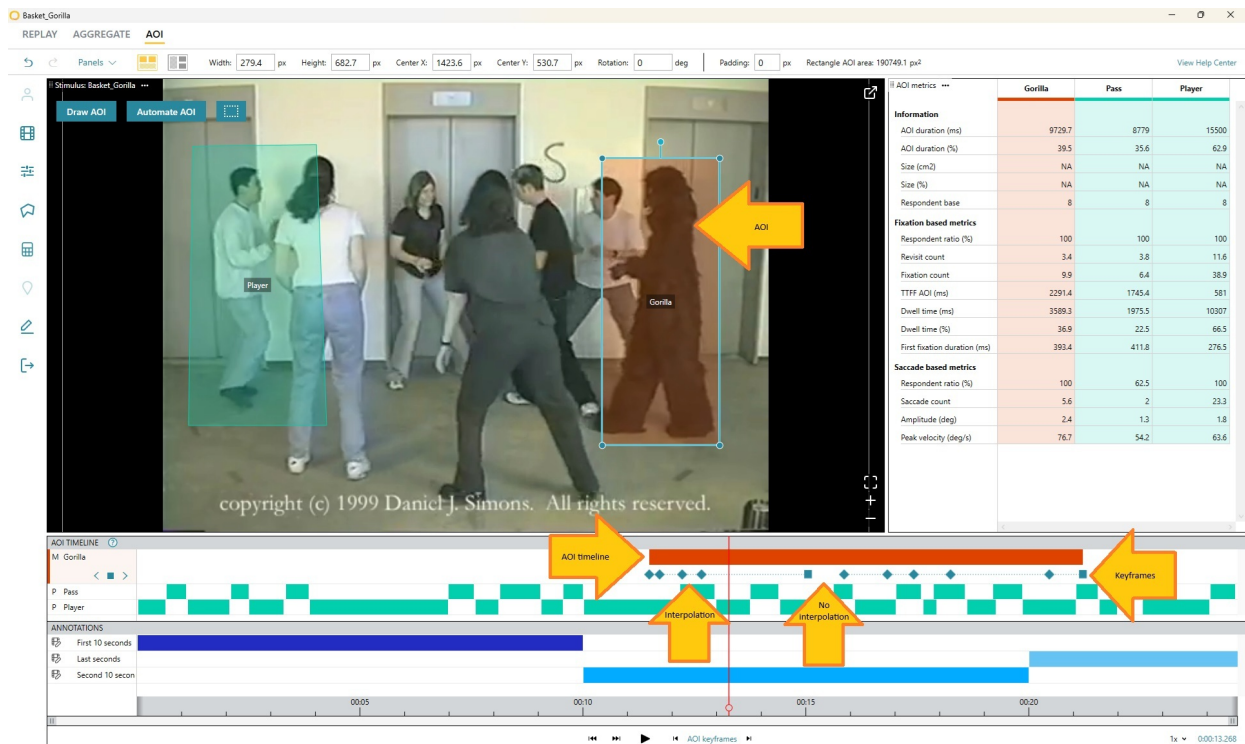
Overview of dynamic AOIs

AOI geometry: As with static stimuli, the AOI is a geometric shape (rectangle, ellipse, or polygon) drawn over the object of interest. If the object changes shape in the stimulus, like becomes bigger or shifts location, the AOI needs to be **reshaped, resized, or relocated**. This can be done manually, or automatically through the Automated AOI feature.

AOI timeline: On dynamic stimuli, the object of interest is not always visible. Consequently, the AOI needs to be active or inactive on the stimulus depending on whether the object is visible. This is represented by the AOI timeline.

Key frames: Every time, the geometric shape of the AOI changes (i.e. when it is made bigger), this is marked with a key frame under the timeline. Key frames are always the result of a user action (reshaping, resizing, relocating) or placed by the automation feature, which updates the AOI geometry at a defined sampling rate.

Interpolation: In between key frames, the AOI will either remain at the location it was placed at, or can be move through linear interpolation. The shape of the key frame as well as the dotted line under the timeline tell you whether an AOI is interpolating in between key frames or not.



Manual options to work with dynamic AOIs

AOI timelines

When you define an AOI on a dynamic stimulus, you will notice that a colored line will appear below the stimulus, starting at the current time point (where the red cursor is placed) until the end of the recording. This is the AOI timeline, not to be confused with Annotations (they look alike and have similar workflows, though).

The AOI timeline indicates when the AOI is active (turned on) and inactive (turned off), corresponding to whether the relevant object is currently visible on the stimulus or not.



There are different ways to set AOI timelines:

- Right after drawing an AOI, an AOI timeline is automatically placed starting at the current frame
- You can manually draw a new interval. When you hover your mouse cursor over the AOI timeline panel, you can see that the cursor changes to a little pen icon. Click and drag to draw an AOI timeline.
- Hit the key on your keyboard that you assigned to your AOI when you drew it

Similarly, there are different ways to change an AOI timeline (see also: [Turning an AOI off and on](#)):

- You can manually drag the start or end of an AOI timeline. When you hover your mouse cursor over the start or end of an AOI timeline, you can see that the cursor changes to a little [or] icon. Click and drag to change the start or end of the AOI timeline. You can also snap the AOI timeline towards the red timeline cursor.

- Right-click on an existing AOI timeline and select "Activate" or "Deactivate" to shorten the interval and make it start or end at the clicked time point.
- Right-click on an existing AOI timeline and select to manually enter a precise timestamp at which you wish your AOI timeline to start or end
- If you want to delete an interval of your AOI timeline, you can right-click on the AOI timeline and select "Delete interval".
- Hit the key on your keyboard that you assigned to your AOI when you drew it

Tip: We recommend always assigning a key to a dynamic AOI. You can do so by selecting a key when you draw an AOI (it is also possible to add a key later, by editing your AOI. Especially if you have to set many AOIs, or have to turn AOIs off and on very often, this is the quickest and most precise way to work.



In case you should accidentally change or delete an AOI timeline, you can always undo this by using the arrow buttons or hitting Ctrl+Z on your keyboard.

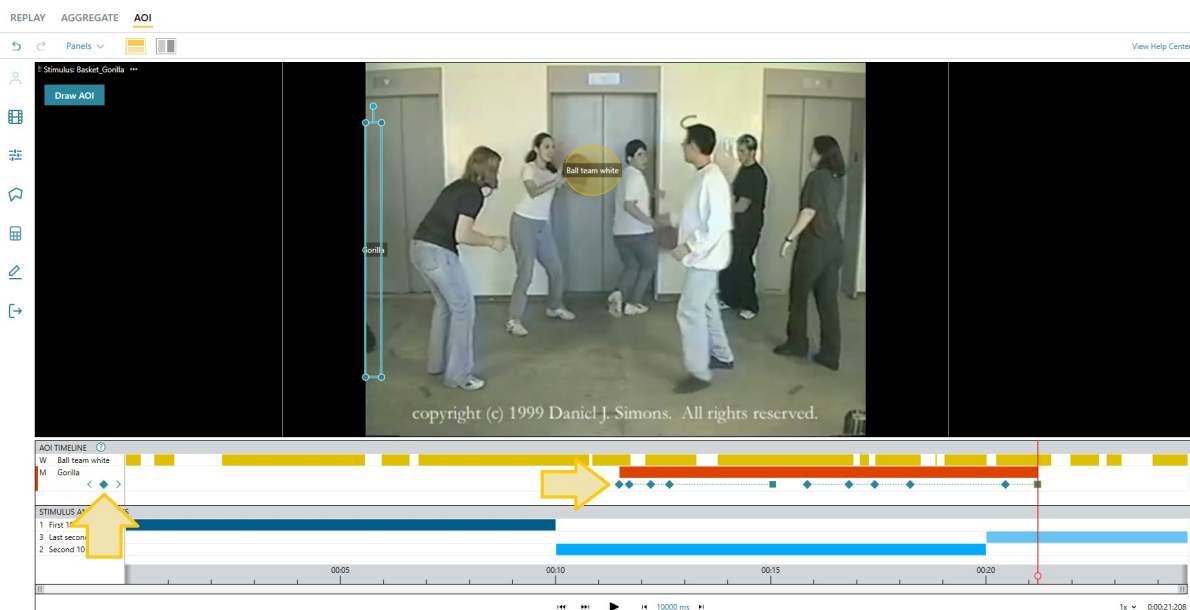
AOI keyframes

Every time you edit the AOI (i.e. change its size, shape, or rotation), an AOI keyframe gets placed. This is indicated by a small square or diamond right below the AOI timeline.

The square indicates an edit at the current keyframe (e.g. change of location), whereas the diamond indicates that the keyframe is also the starting point of a linear interpolation. (Please read more about interpolation in the dedicated section.)

Right next to the AOI name in the AOI timeline panel, you can find a small dialogue to interact with the AOI keyframes < ◆ >

- You can jump between AOI keyframes by using the < and > controls
- Clicking on the square/ diamond will turn interpolation off or on



Turning an AOI off and on

If you have an object that disappears from the screen, you can turn the AOI off by one of the following:

- right-clicking on the AOI shape and selecting "Deactivate"
- dragging the end of the AOI timeline and snapping it to the current frame
- right-clicking on the AOI timeline and selecting "Deactivate"
- hitting the key on your keyboard that you assigned to your AOI when you drew it

While an AOI is turned off, its shape is no longer visible on the stimulus. If you want to turn it back on, you can do so by

- right-clicking on the name of the AOI in the AOI timeline panel and selecting "Activate" (cross-check if this option was introduced!)
- drawing a new AOI timeline towards the red timeline cursor
- right-clicking on the AOI timeline and selecting "Activate"
- hitting the key on your keyboard that you assigned to your AOI when you drew it

Tip: We recommend always assigning a key to a dynamic AOI. You can do so by selecting a key when you **draw an AOI** (it is also possible to add a key later, by **editing your AOI**). Especially if you have to set many AOIs, or have to turn AOIs off and on very often, this is the quickest and most precise way to work.

Please note: Turning an AOI off and on is not the same as **hiding an AOI**. If you hide an AOI, it is just not visible in the Analysis window so that your workspace is less cluttered. Turning it off, however, also means that no AOI metrics will be computed for the time the AOI was turned off.



In case you accidentally turn an AOI on or off, you can always undo this by using the arrow buttons or hitting Ctrl+Z on your keyboard.

Using interpolation

As a default, the AOI will always remain in its current state (i.e. at its current location with the current size) until you edit it. However, especially if you want to track a movement across many frames, you can use linear interpolation to make the AOI move from one location to the other without having to manually adjust it on every frame in between.

You already read about **AOI keyframes** in the previous section. If you edit a dynamic AOI, e.g. change its location, this animation event will be indicated with a little square below the timeline. The AOI will remain at its original location and then suddenly appear at the new location at the indicated keyframe.

If you want your AOI to move from one location to the other (e.g. because you want to track a linear motion), you can activate interpolation. You can either do so by

- clicking on the diamond icon in the dialogue next to the AOI name < ◆ >
- or by right-clicking on the AOI shape and selecting "Interpolate"

If interpolation is turned on at one keyframe, all subsequently placed keyframes will also activate the interpolation. If you want to stop this and do not want your AOI to automatically be moved, you can turn interpolation off by

- clicking on the square icon in the dialogue next to the AOI name < ■ >
- or by right-clicking on the AOI shape and ticking off "Interpolate"

Automated AOIs

Defining and processing automated AOIs

From iMotions 10.1 and onwards, it is possible to place AOIs with the help of computer vision. This feature requires the activation of the Automated AOIs module and works best on PCs with a **NVIDIA GPU of compute capacity between 6.1 and 12.x**. When working with the feature for the first time, iMotions will prompt the user to install the Automated AOIs package. To do so, please follow the steps in the installation wizard.

To define an AOI through the automation feature, you have five options:

- Right click on the object visible in the stimulus and select "Automate"
- Click on "Automate AOI" > New AOI
- If you already have started drawing a dynamic AOI manually (rectangle only), right click on the AOI and select "Automate"
- If you already have an existing dynamic AOI, or defined a dynamic AOI on other respondents, click on "Automate AOI" > Existing AOI

- If you already defined a dynamic AOI on other respondents, click “Automate” on the timeline

Working with Automated AOIs requires the definition of at least one *input frame*, i.e. a frame in the video where the object that you wish to track is clearly visible. You can choose to select more input frames if your object is changing a lot throughout the recording.

On the input frame, you select the object by clicking on it until it is entirely covered by an orange mask. Black dots indicate where you clicked. If the mask exceeds the object, you can *Ctrl+click* to set negative points (red) to tell iMotions that this is not a part of the object.

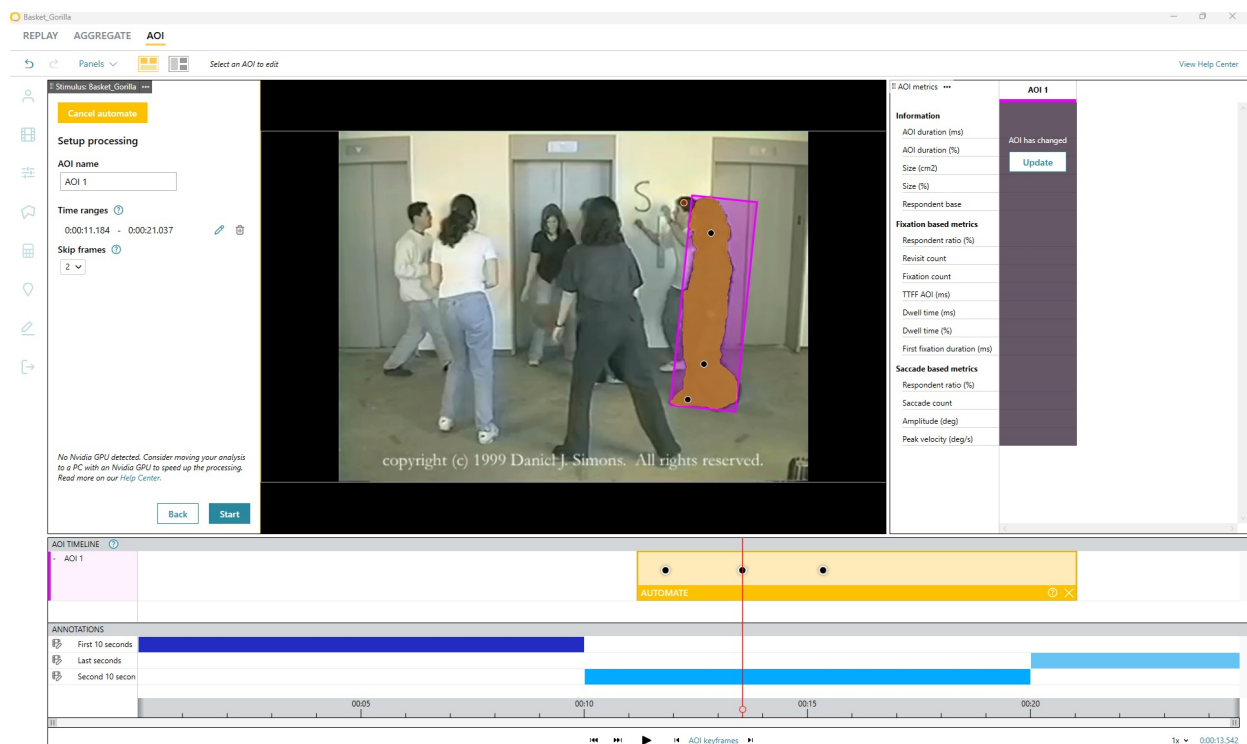
In this step, you can also choose the *Geometry type*, i.e. which shape you want the resulting AOI to be: rectangle, ellipse, or polygon.

The outline around the mask is a preview of the resulting AOI. Please note that dynamic stimuli (videos, recordings) first need to be processed in order to get the actual AOI over the entire stimulus.



For dynamic AOIs, you can see on the timeline (indicated with a black dot) when you set the input frames.

Make sure to select all time ranges during which you expect the object will be visible as only the highlighted times will be processed. If you know that the object is gone during a certain time, it is recommended to exclude this time range as this will lead to shorter calculation times.



Time ranges can be set by

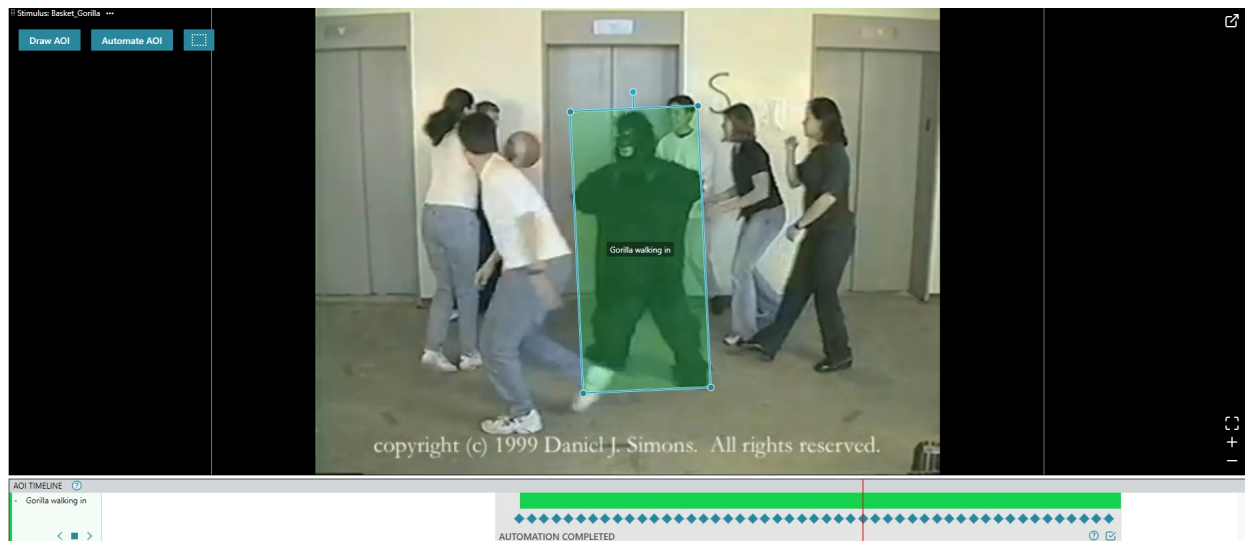
- dragging and drawing them on the timeline,
- right clicking to activate and deactivate,
- or by editing the time stamps of the time ranges in the left side menu.

Lastly, you can choose how many frames of the video you wish to process. By skipping x amount of frames, the processing will go faster and the AOI will be interpolated in between the resulting key frames. Please note that if very many frames are skipped, this can impact the quality of the tracking as the movement will only be coarsely tracked and the object may not be recognized if it changes drastically in between the processed frames.

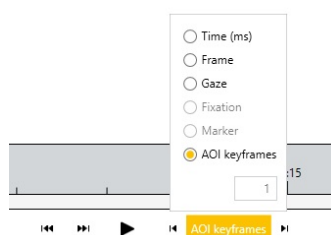


Reviewing and correcting dynamic Automated AOIs

Once the processing is completed, iMotions will return an AOI geometry per processed video frame, indicated by the AOI keyframes on the timeline. Unless you chose to process all frames, i.e. to skip 0 frames ahead, iMotions interpolates the AOI geometry in between key frames.



We strongly recommend you to replay the entire processed time range and visually inspect whether the AOI got placed in a satisfactory way. You can easily step from keyframe to keyframe with the arrow buttons or arrow keys on your keyboard (requires you have an AOI selected). Please note that upon completion of the processing, iMotions will default to stepping per AOI keyframe.

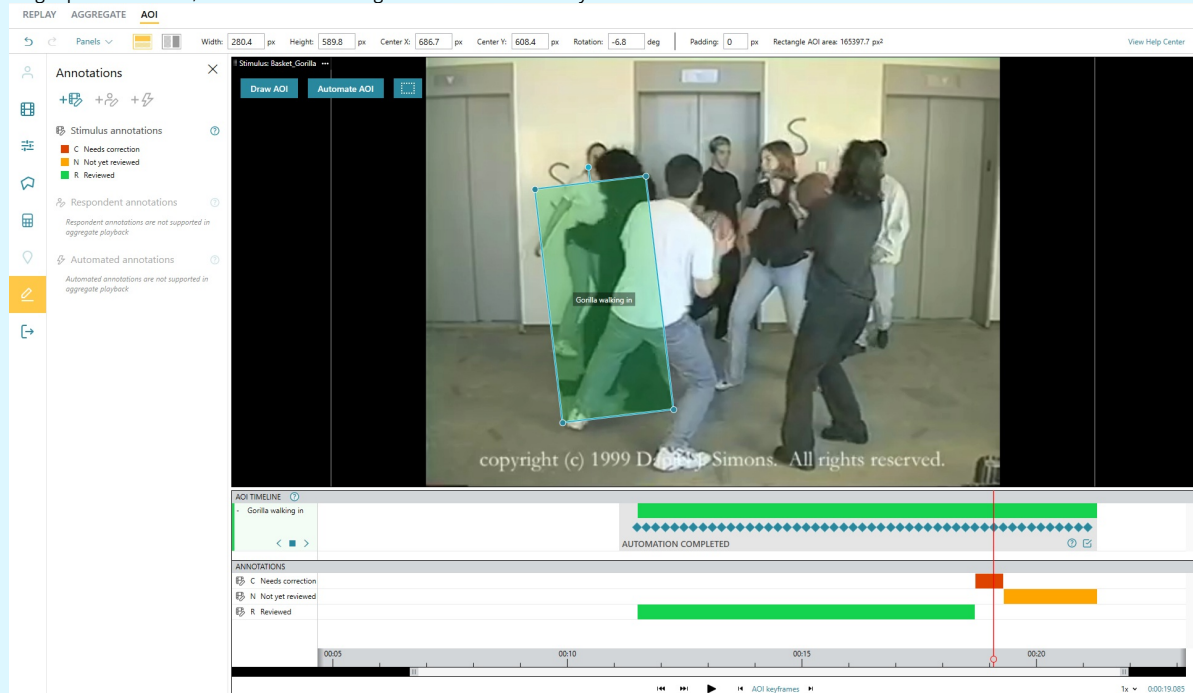


Once you have reviewed the AOI, we recommend you remove the “Automation Completed” message by ticking it off.



Tip

When reviewing longer recordings, we recommend you to make Annotations to mark the time periods that you have reviewed, those periods that you think require further editing, and the time periods that still remain to be reviewed. This can help to keep the overview if working over longer periods of time, or when collaborating with others on the analysis.



If you are not satisfied with the resulting AOI, you have two options:

- manually editing the AOI on the keyframes
- re-running selected time ranges with new input frames

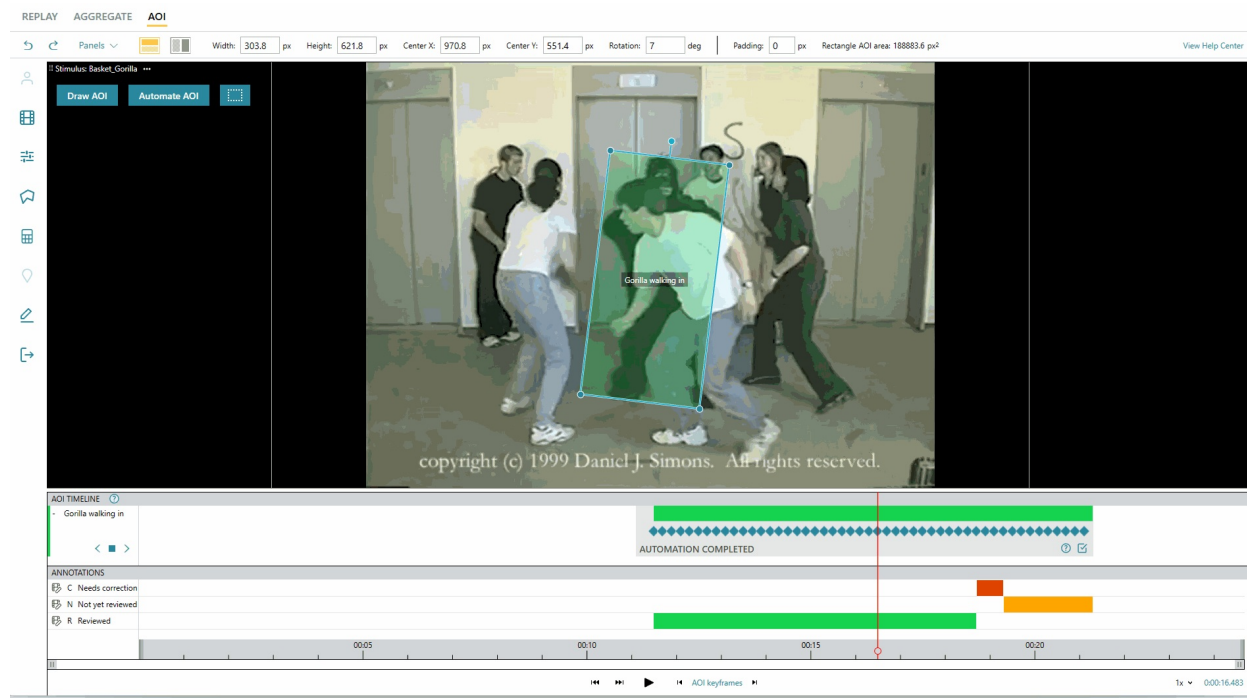
Tip

If you wish to increase the size of the AOI, e.g. to compensate for an inaccurate eye tracker, you can do so by adding padding to the AOI.



To manually correct an AOI, find the key frames where you want to change the AOI and edit it through resizing, rotating, relocating, moving corner points of a polygon, or turning it off or on.

These are the same steps as drawing an AOI manually (see above) and your changes will overwrite the result of the automated processing.



It is also possible to completely clear the AOI timeline as well as key frames in a selected time range by *right clicking on the timeline > Select range to delete..*



If you wish to re-run the automated AOI processing on selected time ranges, this will allow you to provide new input frames and select new time ranges.

The old results in these time ranges (both key frames from the automation as well as manual changes) will be overwritten and replaced with new results. This action can be undone by using the arrow buttons or hitting *Ctrl+Z* on your keyboard.

To reprocess time ranges of an already existing AOI, you can

- click on "Automate AOI" > Existing AOI and then select the AOI
- right click on the AOI and select "Automate"



Tip

When new input frames and time ranges to an AOI that already has an AOI drawn (e.g. from previous processing), then this existing AOI will be shown in black while the preview of the new AOI (the result of the input frames you set) will be shown in color. You can choose to show or hide the existing AOI geometry by right clicking on it.



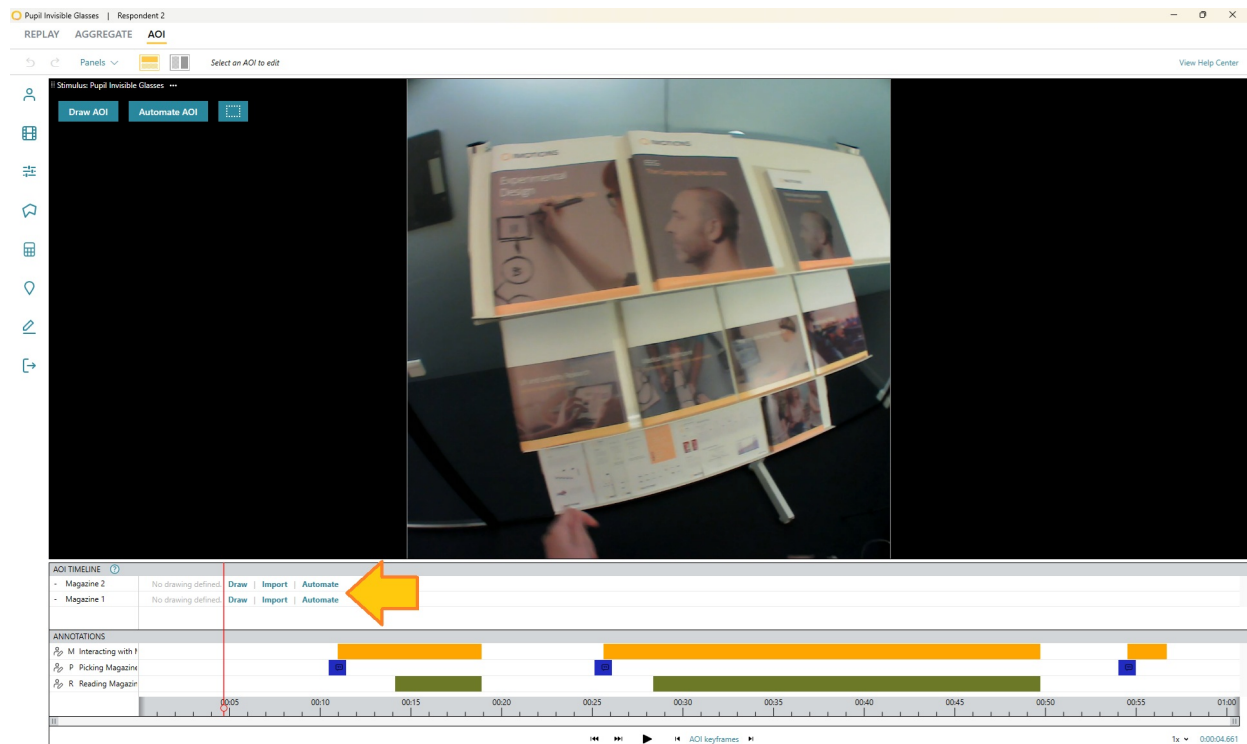
Defining dynamic (manual and automated) AOIs across individual recordings

If you are working with recordings that are individual for every Respondent, such as

- Recordings from mobile eye-tracking glasses
- Recordings from a scene camera setup, including setups using a frame grabber
- Screen recordings
- Website recordings
- VR stimuli

then you will have to define AOIs individually for every single respondent (see also [Best practice tips](#)).

Once you start working on a subsequent respondent, you will see the AOIs you already defined for previous respondents in the AOI timeline panel.



You can choose to

- draw a new AOI shape,
- to import an already existing AOI,
- or to start a new automation for this respondent.

If you choose to import an already existing AOI, you can choose to

- reuse a shape that you already defined for another respondent. This will fetch the first shape that you defined for the other respondent and will import the timeline, i.e. when the AOI was turned on and off. Please note that you will still have to replay the recording and adjust the AOI's size and shape over time.
- import an AOI template

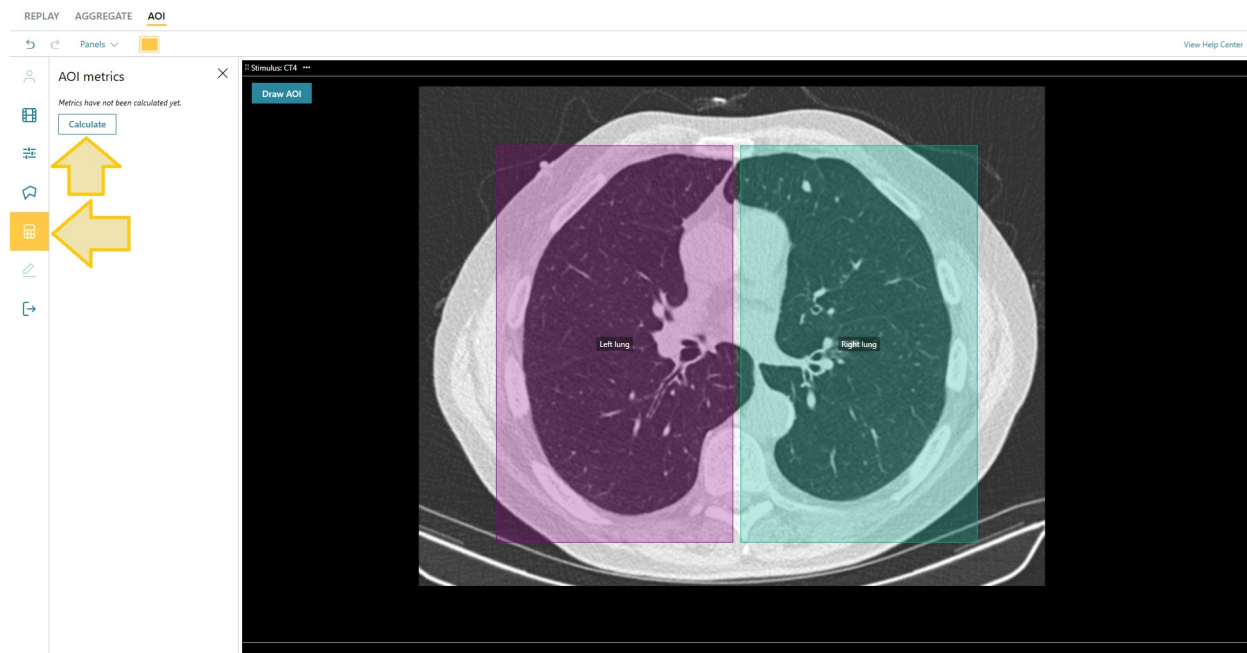
Review AOI metrics

After you have defined at least one AOI, you can review the AOI metrics, which are the eye-tracking statistics that the software automatically computes for you. You can either

- review metrics for the currently visible AOIs in the user interface
- export a table that can be analyzed in statistical analysis software (see section about [metrics and exports](#))

Show AOI table in the UI

Open the AOI metrics menu  and click "Calculate".



Alternatively, if you already have the "AOI metrics" panel open, you can click "Calculate" here. (Remember that you can always customise the panel view using the controls in the top bar.)



Once the AOI metrics are computed (please note: if you have many respondents, many AOIs, a high sampling rate, or all of them then this may take a little while but you can continue working), a table is shown in the "AOI metrics" panel. This table contains results averaged across all respondents in your analysis segment (and for whom you have defined AOIs).

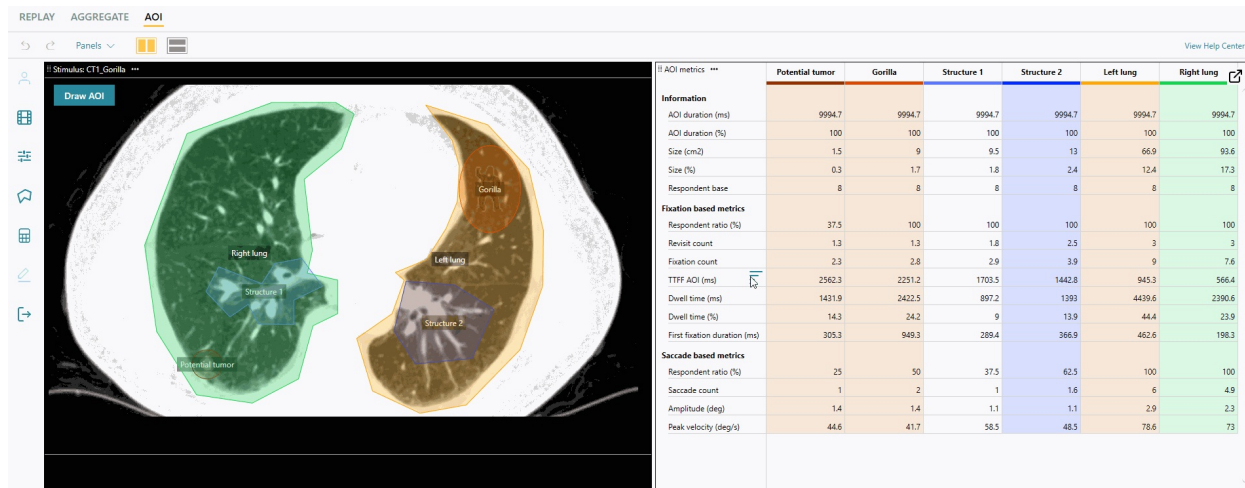
In the [AOI metrics menu](#), you can pick and choose the metrics that you want to visualize in the table and find more information about what the metrics mean.



You can export screen captures of your stimulus with overlaid AOIs along with the AOI table from the **Export** menu on the left-hand side.

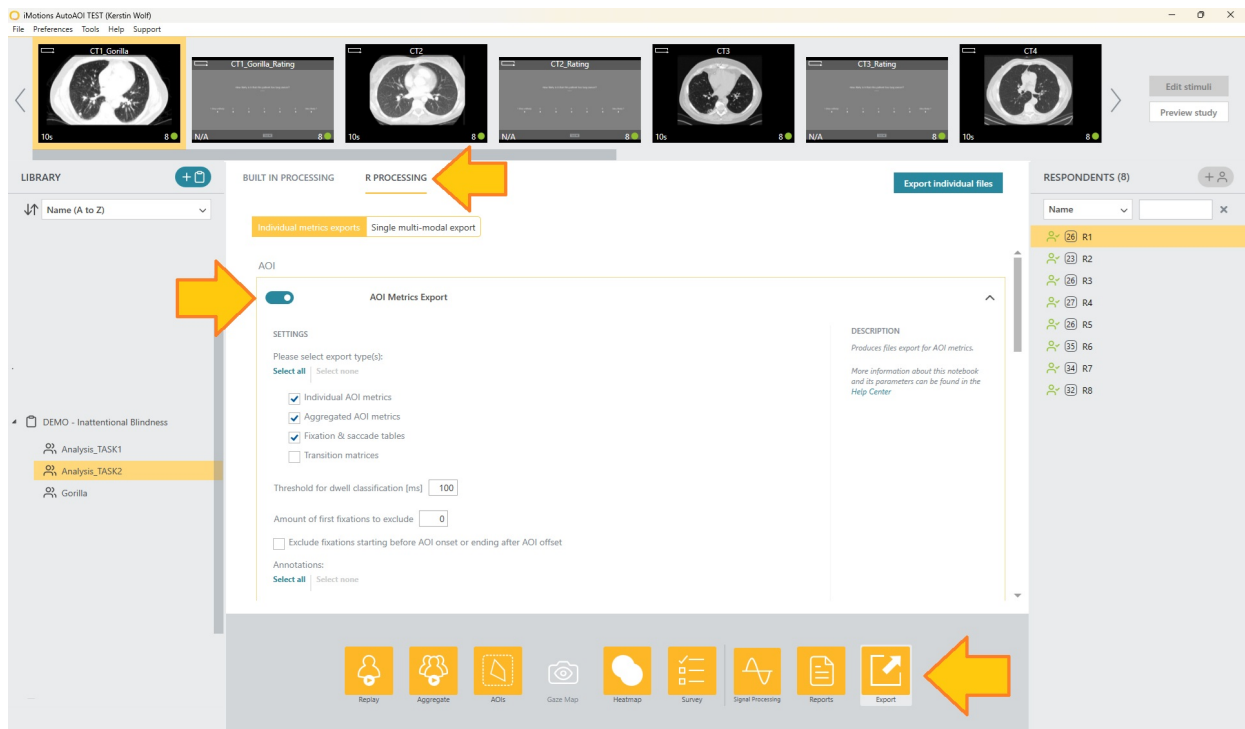
Rank AOIs based on results

You can rank your AOIs based on the results from the AOI metrics computation. For example, if you want to review which AOI was on average looked at the most (longest dwell time) you can choose to sort the AOIs in descending order.



Exporting AOI metrics

For a full overview of all AOI metrics, access to AOI metrics per individual respondent, as well as other types of eye-tracking metrics please refer to the **Export** section in the main UI.



If you wish to export only the selection of metrics previewed in the UI, you can copy them to the clipboard (as comma comma-separated file) and paste them into a spreadsheet editor.

AOI metrics	Left lung	Right lung	Gorilla
Information			
AOI duratic	994.7	9994.7	9994.7
AOI duratic	100	100	100
Size (cm2)	146.5	146.5	10.6
Size (%)	27.1	27.1	2
Respondent base	8	8	8

You can find detailed information about your options in the next section.

Description of metrics and available exports

Basic principles of how iMotions computes AOI metrics

iMotions always presents summarized statistics for different types of eye tracking (and mouse) data. The metrics in the user interface will always be averaged across all respondents in the current segment (for which the AOI was defined; please refer to the "Respondent base" metric to assess how many respondents the metric was averaged).

Metrics categories

Gaze based metrics

Gaze-based metrics refer to the raw eye-tracking data and is based on the sample points derived directly from the eye tracker.

Working with gaze-based metrics can be preferred for data from mobile glasses recordings, in which case the algorithmic classification of fixations and saccades can be disturbed by head movements and reflex-related eye movements. Gaze-based metrics can also be helpful if fixation classification is not possible because of low precision.

Fixation based metrics

Fixation-based metrics refer to data that was classified as fixation by the inbuilt [Duration Dispersion](#), [I-VT filter](#), or [Hidden Markov Models](#).

Working with fixation-based metrics is generally preferred because it allows inferences about the actual intake of visual information. After all, the eyeball needs to stand relatively still to process visual information on the fovea.

Saccade based metrics

Saccade-based metrics refer to data that was classified as a saccade by the inbuilt [I-VT filter](#) or [Hidden Markov Models](#). Saccades are very fast, ballistic eye movements transitioning the gaze from one location to the other. Saccade metrics tell more about the redirection of attention as well as reading patterns.

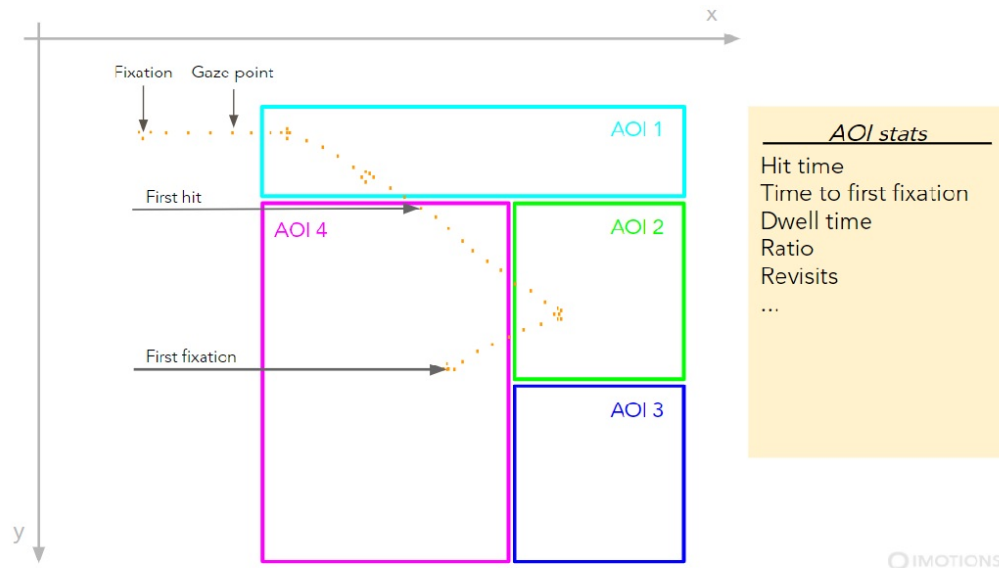
Mouse based metrics

iMotions can count the number of clicks inside an AOI.

Processing steps

Raw eye tracking is filtered by the selected fixation classification filter. You can read more about the available filters, the [Duration Dispersion](#), [I-VT filter](#), or [Hidden Markov Models](#), on our [Help Center](#).

All AOI metrics are primarily based on gaze dwells, which means that gaze samples were recorded inside or outside an AOI.



This dwell classification is converted into a binary signal classifying the gaze data as "inside" or "outside" an AOI. If a gaze sample falls exactly onto the bounding box of an AOI, it is classified as falling "inside" the AOI.

Both blinks and data dropouts can lead to missing data. If the gaze samples before and after the missing data fall inside the same AOI, the entire period of missing data is counted as belonging to the same dwell.

To compensate for outliers in the raw data (single samples that fall "on the other side" of the AOI boundary), a threshold can be applied that first classifies a dwell after gaze has been recorded inside the AOI for the specified time window. Likewise, a dwell will first be classified as ended once the gaze has left the AOI for the specified time window. This *threshold for dwell classification* is by default set to 100 ms but can be changed when AOI metrics are exported.

To avoid biases in the AOI metrics arising from the gaze direction during the preceding stimulus, it is possible to exclude fixations that happened right after Stimulus onset (or the onset of an equivalent time interval, for example, a Gaze Map). By default, all fixations are included in the metrics, but the *amount of first fixations to exclude* can be changed when AOI metrics are exported.

Since fixations can start before the onset of an AOI, Stimulus, or other equivalent time interval (and likewise, end after the AOI's offset et cetera), you can choose to discard partial fixations. By default, a fixation that starts before or ends after the AOI gets truncated so that the metrics compute it as starting at the start of the AOI, or ending at the end of the AOI. If you wish to *exclude fixations starting before AOI onset/ ending after AOI onset*, you can do so when exporting AOI metrics.

After all fixations have been classified and processed according to the parameters described above, iMotions computes which AOI these fixations fell inside. This is based on the binary classification of gaze samples falling "inside" or "outside" the AOIs. A fixation is classified as falling "inside" an AOI if at least one gaze sample was classified as "inside" an AOI. This can lead to a fixation falling right at the edge of two neighboring AOIs get classified as belonging to both AOIs. Likewise, if a fixation remains static on the computer screen but an animated AOI briefly moves over that fixated location so that they overlap for at least the duration set as the *threshold for dwell classification*, the fixation will be classified as having fallen "inside" that AOI.

Based on this classification of raw gaze data falling "inside" or "outside" an AOI, and likewise fixations falling "inside" or "outside" an AOI, all gaze and fixation-based metrics such as Hit time, Time to First Fixation, Revisits et cetera are classified.

For saccade-based metrics, entry saccades are classified as the first saccade during which the gaze transitioned from a location outside the AOI into the AOI. Exit saccades work the other way round, they are the last saccades leaving the AOI. All other saccade-based metrics refer to saccades that start and end inside the AOI's boundaries. Since saccades are very fast events, they can be missed by the algorithm (especially when data is recorded at a low sampling rate). In this case, the saccades metrics show NA values.

Export options

All eye-tracking metrics can be exported for further statistical analysis. Different exports contain information about eye movement behavior and AOIs:

- The **Sensor Data export** can be configured to contain the raw eye-tracking data as well as processed eye-tracking data from the fixation classification algorithms. You can also add markers indicating when AOIs were placed on the stimulus as well as when they were being gazed at.
- If you wish to export the **AOI summary metrics** that you can also review in the user interface, you can choose to export **Aggregated AOI metrics**. For statistical comparison, we recommend exporting the **Individual AOI metrics**, which summarize each respondent's viewing behavior on the AOIs rather than averaging the AOI metrics across the entire segment.

- If you wish to compute your own metrics based on the detected oculomotor events, you can export *fixation and saccade tables*, listing all classified fixations as well as saccades ordered by their time of occurrence, along with their associated metrics. A separate table is generated for Scenes and Gaze Maps because the remapping of gaze samples to the reference image can lead to an altered distribution of samples. Fixations and saccades are classified according to the same parameters as for the original recording so the validity of this classification needs to be evaluated with regards to the setup, recording conditions, and resolution of the reference image.

- To understand more about transitions between the different AOIs (a transition is defined as two subsequent fixations falling into different AOIs), you can export transition matrices. A transition matrix lists all AOIs that were present during the selected interval (Stimulus, Scene/ Gaze Mapping, Annotation) and presents the count of transitions that originated in the AOI shown in the rows and ended in the AOI shown in the columns. iMotions will generate one folder per interval, by default only for all Stimuli and Scenes/ Gaze Mappings (Annotations can be included using the *Create transition matrices based on Annotations* option) and one file per respondent, with an additional file showing the aggregation, i.e. the sum of all transitions across all respondents.

	Paragraph 1	Paragraph 2	Paragraph 3	Question 1	Question 2	Question 3
Paragraph 1	NA	3	1	2	1	0
Paragraph 2	4	NA	1	1	0	0
Paragraph 3	1	1	NA	1	2	1
Question 1	2	1	1	NA	2	0
Question 2	1	0	1	1	NA	4
Question 3	1	1	2	0	1	NA

When exporting your data, you can change the following parameters:

- Threshold for dwell classification
- Amount of first fixations to exclude
- Exclude fixations starting before AOI onset/ ending after AOI offset

Please refer to the [previous section](#) to learn more about their meaning and impact.

For *Individual* and *Aggregated AOI metrics*, you can choose to include Annotations in AOI metrics. In addition to receiving AOI metrics for the entire stimulus duration, you will then also receive a breakdown of AOI metrics for the time windows during which an Annotation was active on the stimulus. (If an Annotation consists of multiple intervals, the AOI metrics will not be further broken down by interval.)

For fixation and saccade tables, you can also choose to include Annotation and Scene markers, which will indicate which Annotations and Scenes were active during every listed oculomotor event.

Introduction to AOI analyses

About working with AOIs

With AOIs, you can define where and when those objects appeared that matter for your study, and for which you want to analyze if a respondent saw them or not. Since the eye tracker only knows the direction in which the respondent was looking, but not which object they looked at, you have to set AOIs around the objects that are relevant for your analysis so that the software can analyze when the respondent looked at the objects.

This has both a spatial component (where on the screen is the object?) as well as a temporal component (when was the object visible?). The latter only applies to dynamic stimuli such as videos, website recordings, screen recordings, or studies with mobile eye-tracking glasses, in which case objects can appear and disappear over time.

Once AOIs are defined, this allows different kinds of analyses.

- Firstly, it allows to infer the visual salience of the objects inside the AOI, e.g. because respondents quickly directed their gaze onto the AOI once the object appeared in their field of view, as well as the objects' complexity or relevance for the task, e.g. because respondents dwelled on them for a longer time than other comparable objects. This can be easily analyzed from iMotions' exports.
- Secondly, AOIs allow for scan path analyses, which means analyzing the order in which objects were being looked at and analyzing the similarity of content exploration between respondents. This allows us to infer whether the stimulus caused similar thought processes in the respondents, which led them to redirect their visual attention in similar ways. These types of analyses require further statistical processing, but the basics for these analyses are provided from iMotions' exports.

For a thorough introduction to eye-tracking analyses, we recommend reading Holmqvist et al. (2011). *Eye Tracking : A Comprehensive Guide to Methods and Measures*. Oxford University Press.

AOI metrics vs. other eye-tracking metrics

iMotions offers you different options to work with eye-tracking data:

1. You can review and export *raw eye tracking data*, usually referred to as gaze data. This data can be reviewed as graphs in the [Replay editor](#), and exported as Sensor Data.
2. This raw eye-tracking data can be processed to detect oculomotor events such as *fixations and saccades* with the inbuilt [I-VT](#), [Hidden Markov Models](#), or [Duration Dispersion](#) fixation classification filters. Fixation and saccade metrics can also be reviewed in the [Replay editor](#), and exported along with the raw Sensor Data from other sensors. For dedicated analyses of fixation and saccade metrics as well as scan paths, you can export [fixation and saccade tables](#) from iMotions.
3. Raw eye-tracking data can also be used to [detect blinks](#).
4. By defining AOIs, you can quantify the amount of raw gaze data or the number of oculomotor events on an object that matters for your study. *AOI metrics* can be reviewed in iMotions user interface (as an average of all respondents of the selected segment), or exported for further statistical analysis
5. *Transition matrices* are used to understand the probability that the respondent shifted their visual attention from one AOI to the next.

AOI editor vs. other editors

When you are working in the Analysis window, you can see tabs that represent the different data editors:

- The purpose of the AOI editor is to draw AOIs according to your research objective and to quickly review AOI metrics for the current stimulus.
- You can review an individual respondent's data, both the eye tracking data as well as data from other sensors, in the [Replay editor](#).
- The [Aggregate editor](#) allows you to review the aggregated gaze data from all respondents simultaneously, along with aggregated signals. Please note that data aggregation is only possible if the stimuli are static (Images, Survey slides, [Gaze Maps](#), [Scenes](#)) or if the stimulus has the same timeline for all respondents, like for Videos.

AOIs that are set for those stimuli that can be [aggregated](#) are automatically applied for all respondents simultaneously, so you only need to define AOIs once per stimulus. Certain stimuli, however, cannot be aggregated because their recordings are unique for every respondent. This is the case for recordings from mobile eye-tracking glasses or scene camera setups, screen recordings, website recordings, and VR stimuli. In this case, you will have to define AOIs for every individual respondent.

Accordingly, all gaze visualizations in the AOI editor are by default [aggregated](#), unless you are working on an individual recording. If you want to review an individual respondent's gaze data along with the AOIs on a static stimulus or video, then you can do so in the [Replay editor](#).

The three editors are synchronized with each other. All AOIs that you define in the AOI editor can also be viewed in the [Replay](#) and [Aggregate editor](#), for inspection of the individual or aggregated data along with the AOIs. AOI metrics can only be reviewed in the AOI editor.

If you are working with Annotations, you can set Respondent Annotations in the [Replay editor](#) and Stimulus Annotations in the [Aggregate editor](#). Annotations are synchronized with the AOI editor as well; for static stimuli and videos, you have access to Stimulus Annotations that are synchronized with the [Aggregate editor](#), and for individual recordings, you have access to Respondent Annotations that are synchronized with the [Replay editor](#).

Best practice tips

Plan your AOIs already when designing the study

When starting your study and defining your research question, ask yourself how you want to analyze your data, which statistical comparisons you want to run, and which AOIs you will need to define to answer your research question. Formulate a hypothesis about the objects you draw your AOIs around, and decide beforehand which AOI metrics you will analyze to accept or reject your hypothesis.

It can be tempting to draw lots of AOIs once the data has been collected, but this can lead to irrelevant findings because statistically speaking, there will always be some significant differences as long as you do many comparisons, but they will not necessarily be replicable.

Also, defining more AOIs than really needed will increase the manual work and time of your analysis. This is particularly relevant the more natural your study was, for example during screen or mobile eye tracking glasses recordings.

Manual effort and analysis time when drawing AOIs

Drawing AOIs on a static stimulus is easy. Since the stimulus does not change over time, you do not need to [animate your AOIs](#), and because static stimuli are always the same for all respondents, they can be [aggregated](#) and the AOI only needs to be placed once rather than drawn it for every single respondent. Of course, the manual work and therefore analysis time increases if you have many AOIs or many stimuli, but in this case, you can use [templates or duplication](#) to make your analysis more efficient.

When working with videos, the effort of working with AOIs increases. Because the objects in the video change their location, disappear, or reappear, you will have to [animate your AOI](#) to track the objects. Here, your analysis time can be reduced by using [interpolation](#). The good thing with videos is that they can be [aggregated](#) and your AOIs are automatically defined for all respondents.

If your stimuli contained recordings, e.g. from mobile eye-tracking glasses, then your analysis time will scale quickly. Recordings are always unique for every respondent, and it depends on the respondent's behavior during the experiment at which time and at which exact location the objects that you want to draw an AOI on appear in the recording. As an example, if you observe shoppers in a supermarket, they will walk at their own pace and review different sections of the supermarket before purchasing the product. Similarly, if you observe an interaction with a mobile phone, people will scroll at their own pace. Therefore, you will not only have to [animate your AOI](#), but also go through every respondent's recording and set AOIs individually. The more respondents you have in your study, the longer your analysis will take and usually, you cannot use templates because the recordings will be so unique.

[Automating the AOI placement](#) is a good way to save time, but requires review and likely also corrections.

As a general best practice, we recommend planning your AOIs already in the design phase of your study, using the simplest possible [AOI shape](#), and considering [Gaze Mapping](#) where possible. If you are running your analyses under high temporal pressure, we recommend that you run a pilot study with at least one respondent and use this pilot data to define AOIs. This allows you to either save [templates](#) that you can then quickly apply to your actual study, or to just record data from all "real" respondents in the same study and afterward exclude the pilot respondent(s) from your statistics.

Consider Gaze Mapping or Scenes to simplify AOI analysis

For studies that contain recordings, for example from mobile eye tracking glasses, websites, screen recordings, or after testing on mobile phones and tablets, there are ways to make your analysis easier. As discussed above, these types of stimuli cannot be [aggregated](#) because the recordings are unique per respondent.

If every respondent saw the same thing (like a shelf, a website, an app's user interface, et cetera) but explored it at their own pace, could walk towards it or scroll on it, then it is possible to use an extra step to create a new visualization from the original recording, on which the individual respondents' data can be aggregated. Depending on your type of study, you can do so by creating [Web Scenes](#) or [Gaze Maps](#).

This extra step will produce a new image, on which it is easy and efficient to draw AOIs.

Choosing the right AOI size and shape

When drawing AOIs, one must be aware that no eye tracker is perfect. There is always some measurement error and therefore, AOIs should ideally include some padding around the actual object. This can be added manually by drawing the AOI bigger, or by adding it as an AOI parameter. The amount of padding as well as the distance between neighboring AOIs influences the amount of false positives (the respondent looked at another object, but their gaze is counted as part of the target AOI) and false negatives (the respondent looked at the target object, but since the eye tracker was off and the AOI too small, their gaze is not recorded on the target AOI).

Although it may be tempting to draw very precise AOIs with the [polygon](#) tool, it is often better to use a more simple shape, firstly because this automatically adds padding around the AOI and secondly because they are easier to draw, [animate](#), and [edit](#).

Choosing the ideal AOI size depends on different factors. The most critical factor is the accuracy and precision of the used eye tracker together with the quality of the calibration, but also the accuracy of the respondent's saccadic eye movements as well as the layout of the stimulus ([Orquin et al., 2015](#)). The worse the eye tracker, the more padding you should add. [Holmqvist et al. \(2011\)](#) recommend adding around 1-1.5 dva padding when placing an AOI. (For screen-based eye tracking setups, this means a margin of circa 1-2 cm.) Beyond this margin, the authors recommend placing AOIs as precisely as possible. Please be aware that some consumer eye trackers may require larger margins, and webcam eye tracking requires an even greater buffer.

[Goldberg and Helfman \(2010\)](#) recommend considering the task that the respondent performed and the importance of *every* fixation being captured (reduction of false negatives), whether the stimulus is crowded or if the target object is surrounded by white space, and how much variance in fixation positions the researcher expects. If a stimulus is not crowded, meaning the object of interest is mostly surrounded by homogeneous space, [Hooge and Camps \(2013\)](#) conclude that the AOIs should be as large as possible.

References and further readings

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